

BFS Traversal

Step 1: Start at E

Legend

Complete

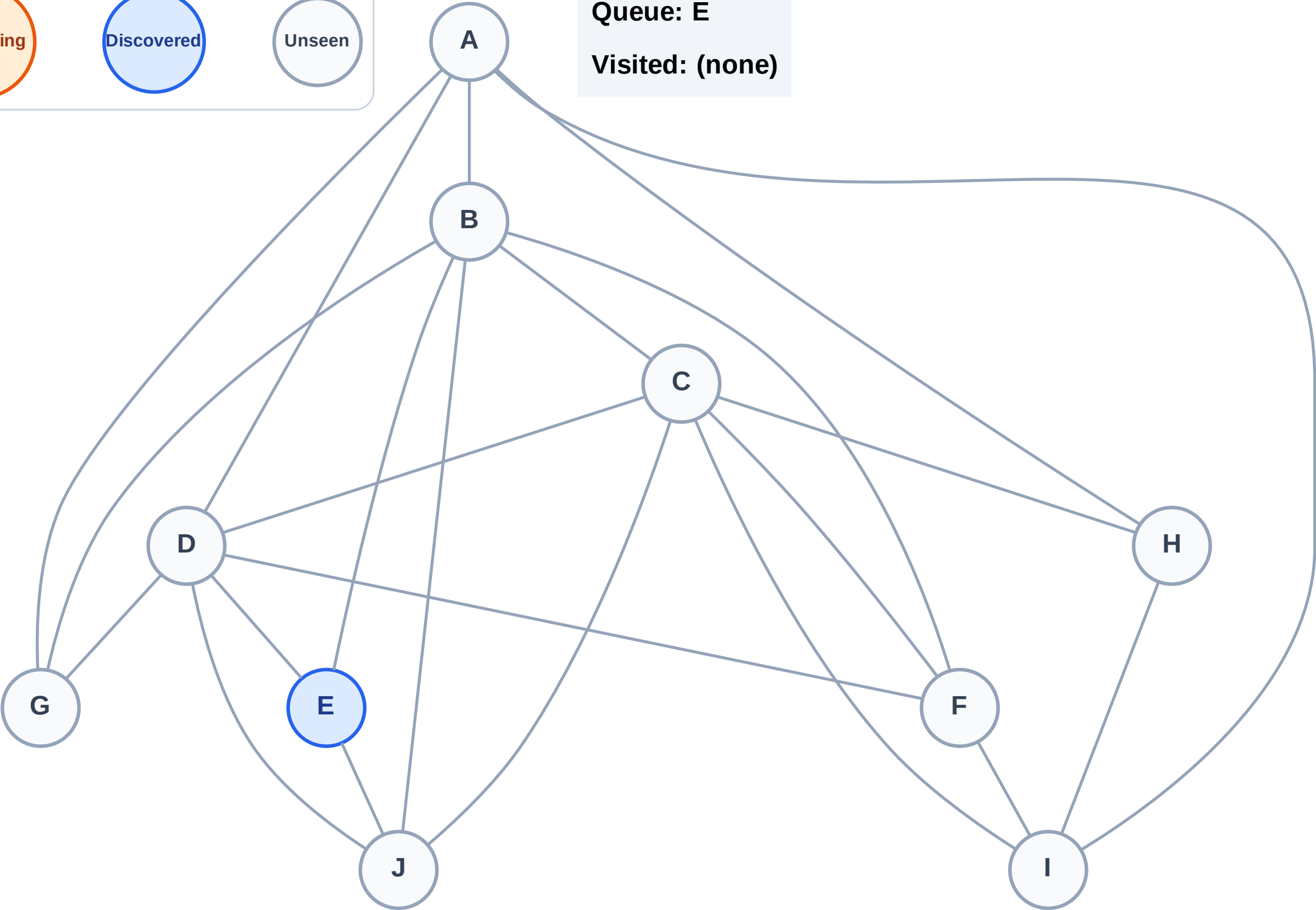
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

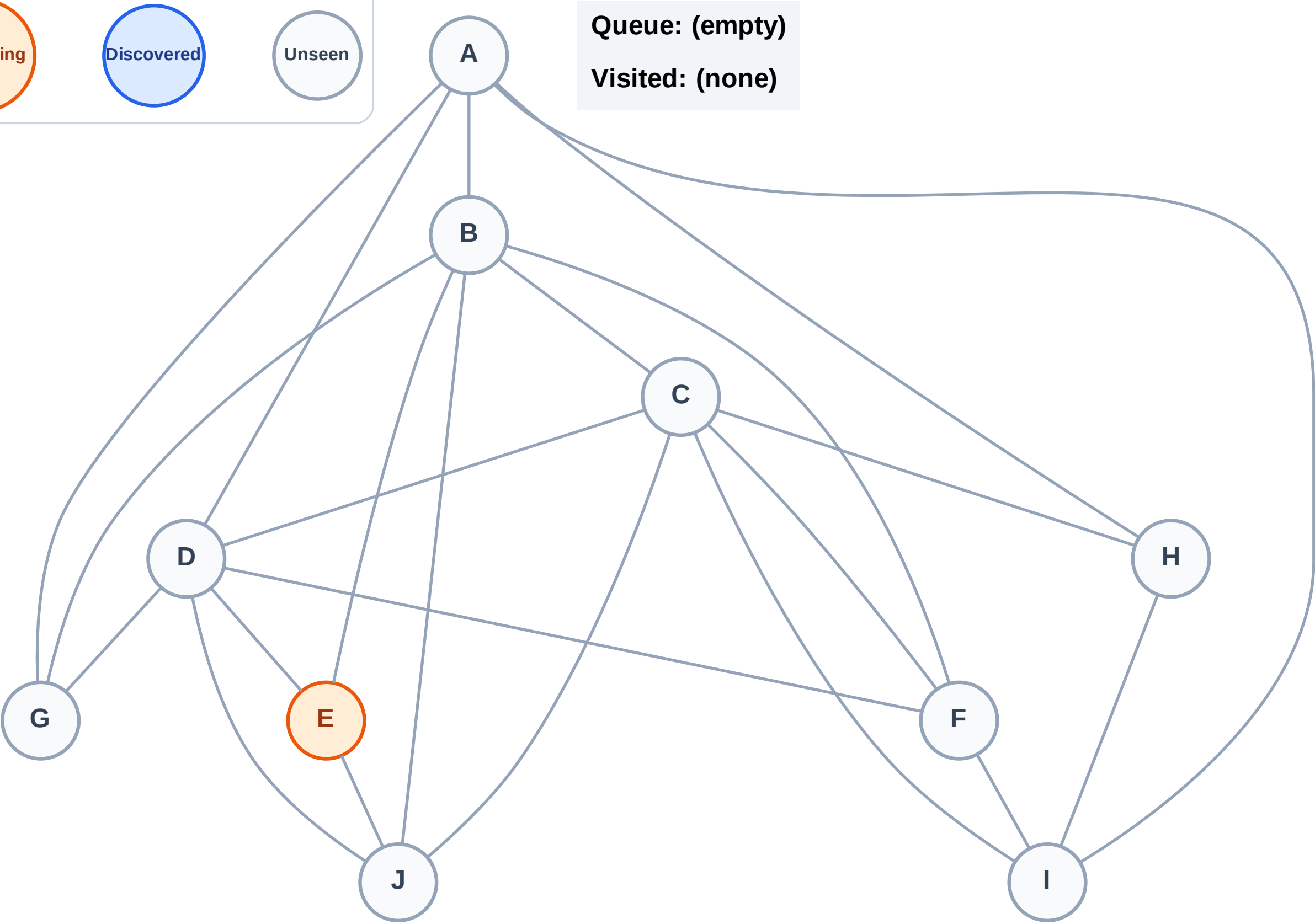
Complete

Processing

Discovered

Unseen

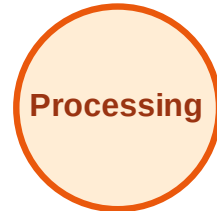
Queue: (empty)
Visited: (none)



BFS Traversal

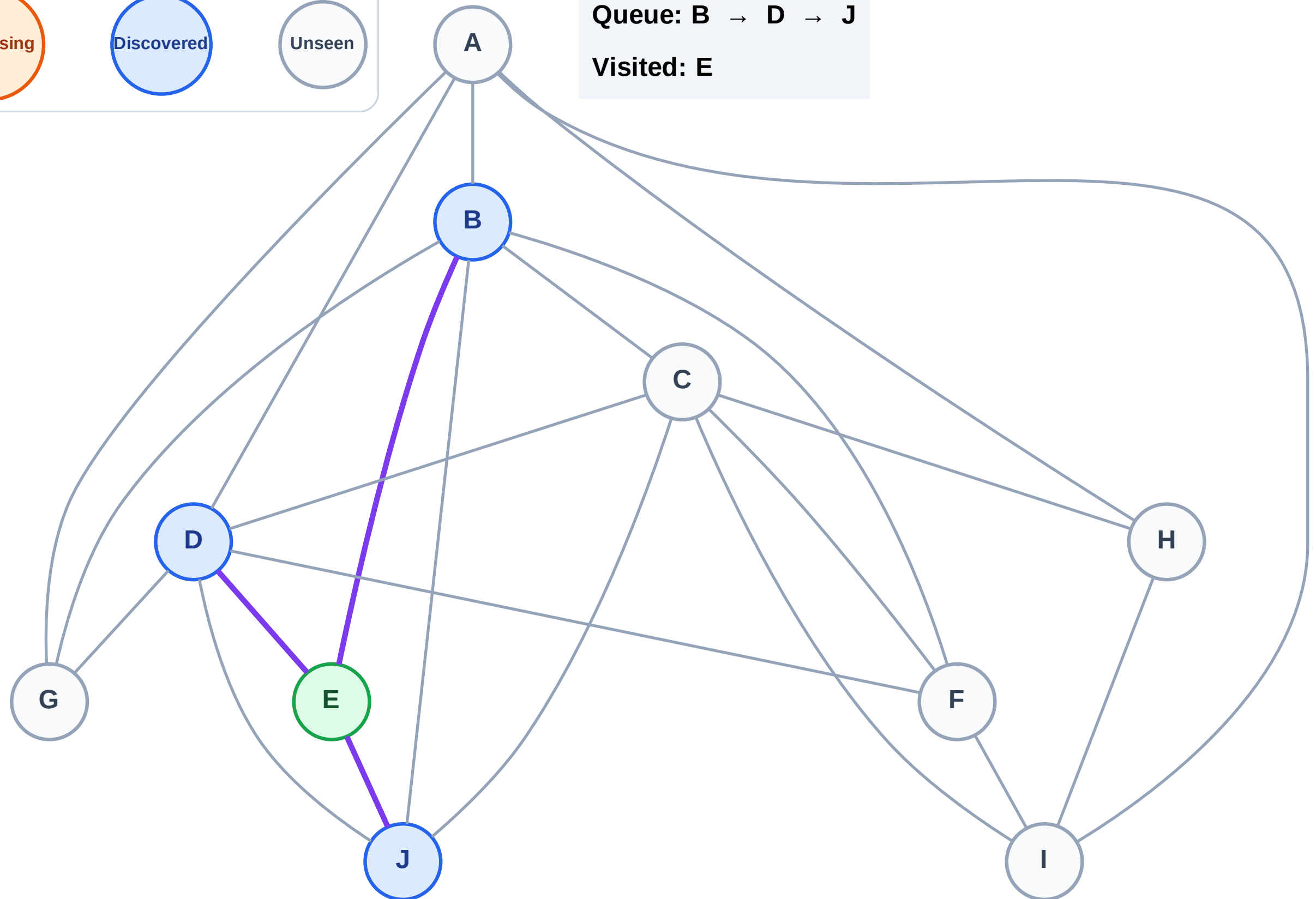
Step 3: Discover B, D, J from E

Legend



Queue: B → D → J

Visited: E



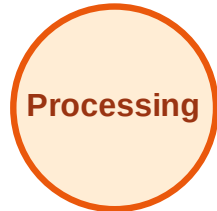
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



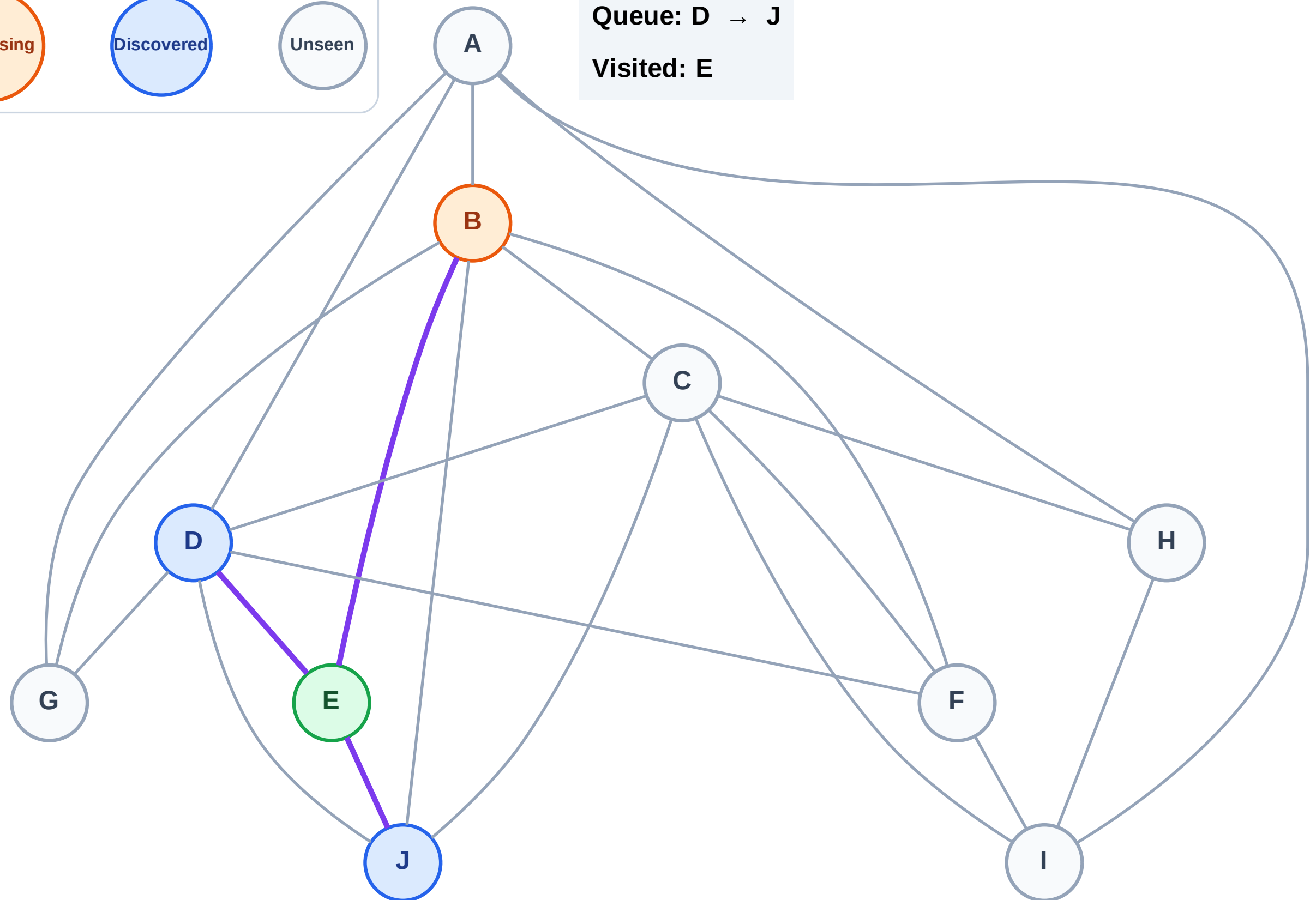
Discovered



Unseen

Queue: D → J

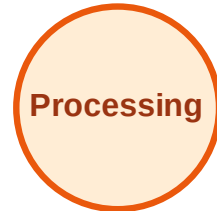
Visited: E



BFS Traversal

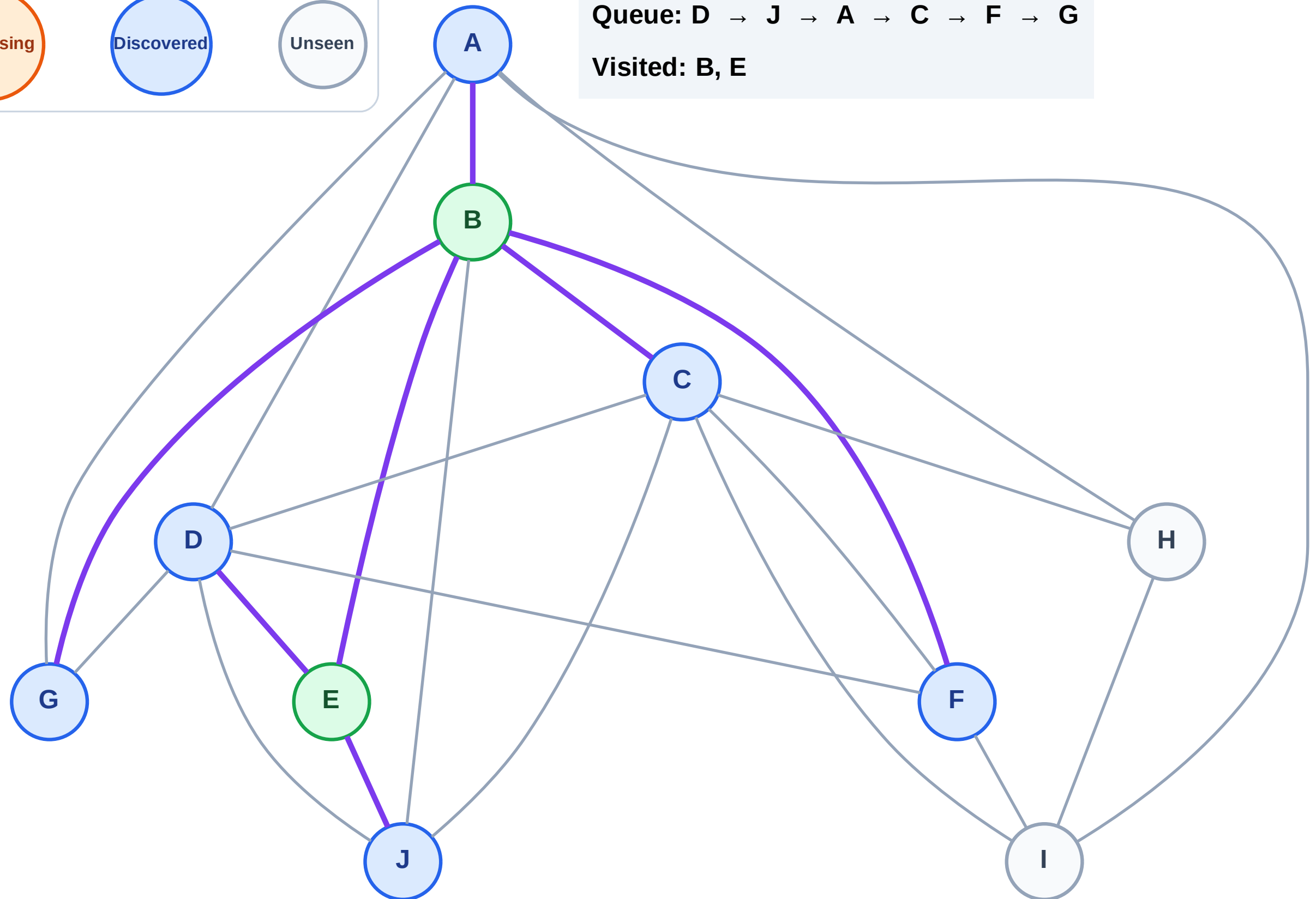
Step 5: Discover A, C, F, G from B

Legend



Queue: D → J → A → C → F → G

Visited: B, E



BFS Traversal

Step 6: Process node D

Legend

Complete

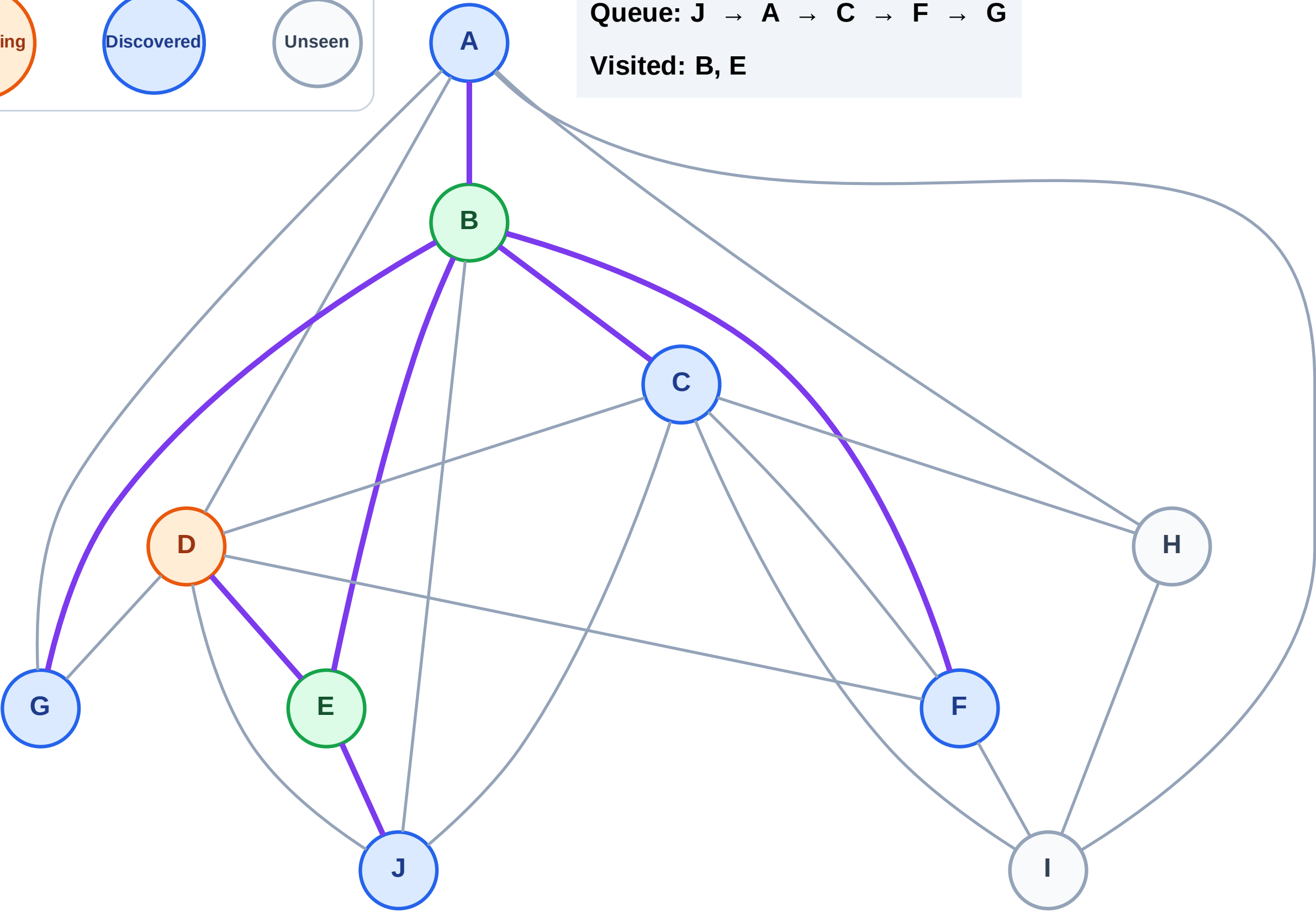
Processing

Discovered

Unseen

Queue: J → A → C → F → G

Visited: B, E



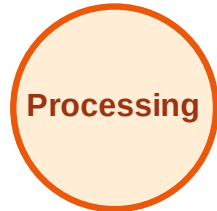
BFS Traversal

Step 7: No new neighbors from D

Legend



Complete



Processing



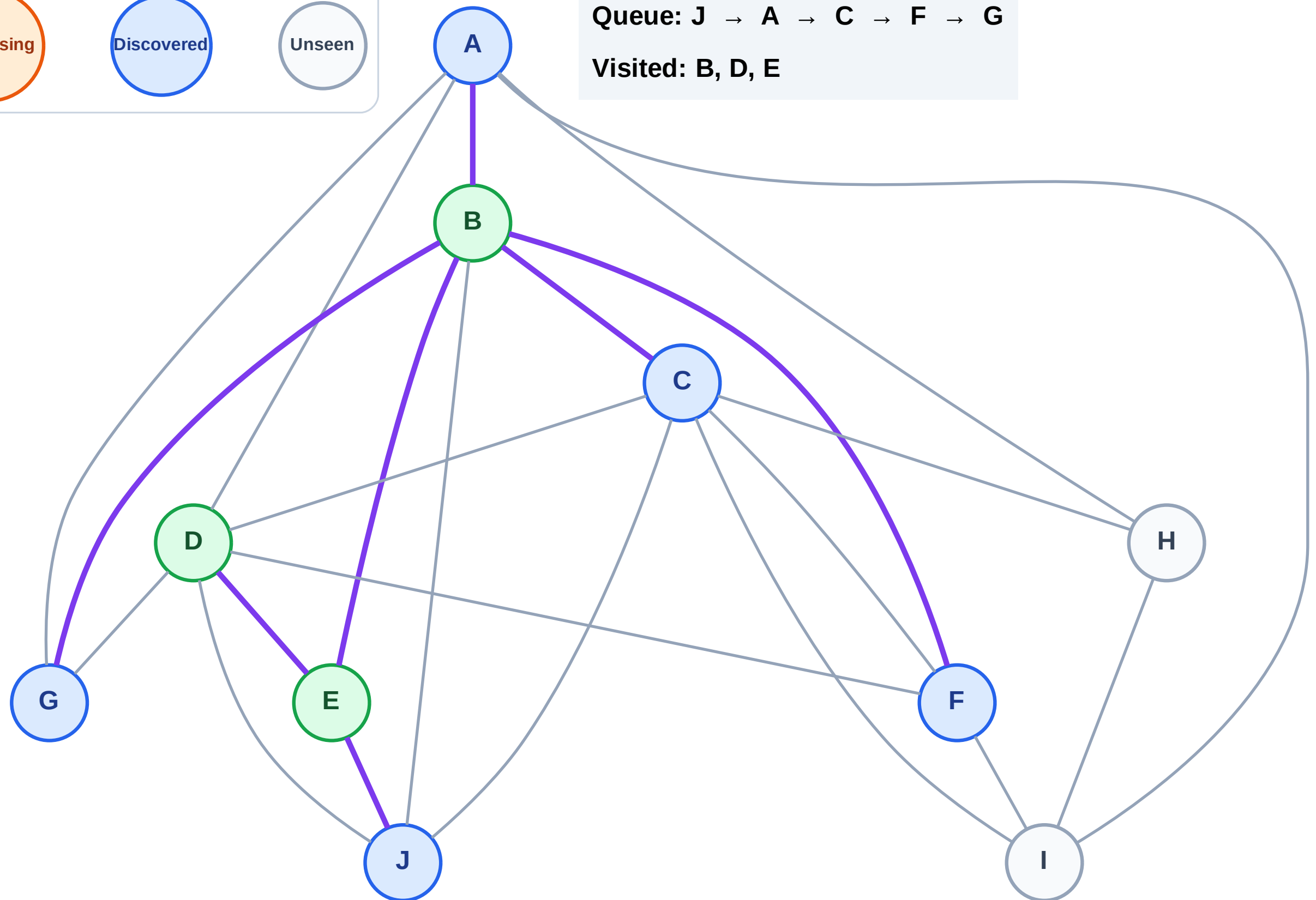
Discovered



Unseen

Queue: J → A → C → F → G

Visited: B, D, E



BFS Traversal

Step 8: Process node J

Legend

Complete

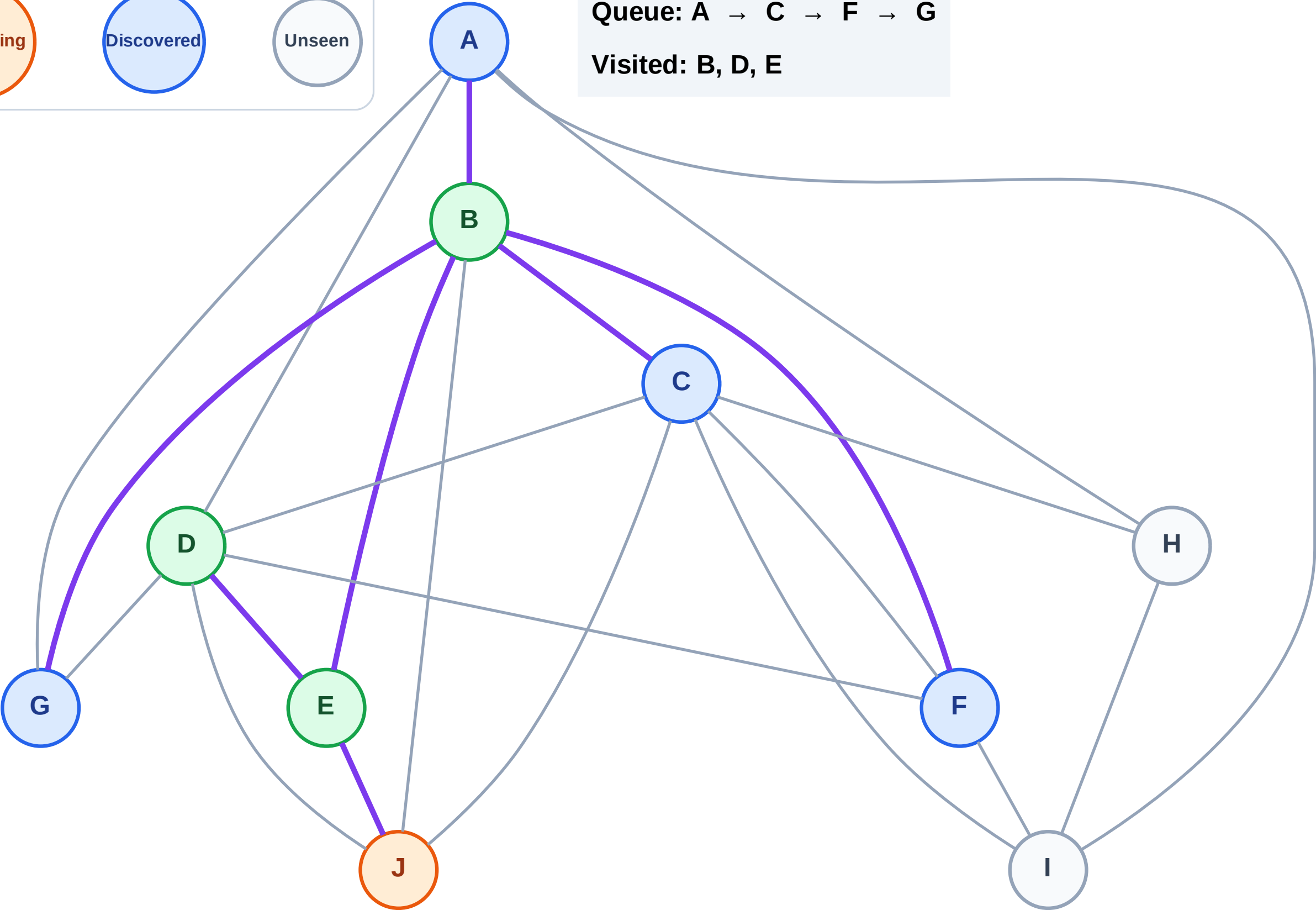
Processing

Discovered

Unseen

Queue: A → C → F → G

Visited: B, D, E



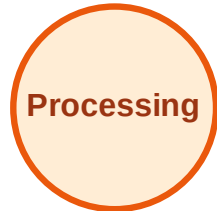
BFS Traversal

Step 9: No new neighbors from J

Legend



Complete



Processing



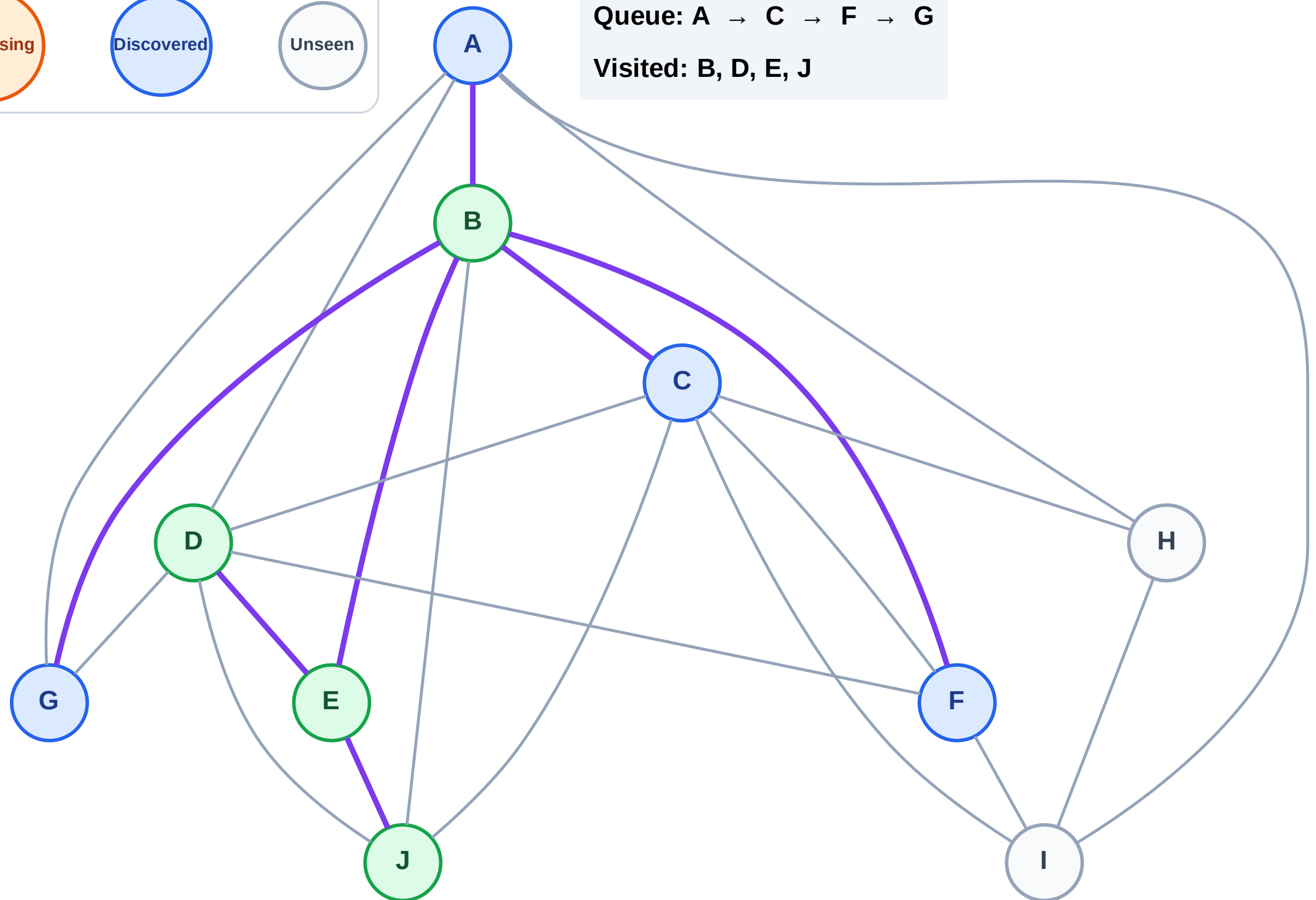
Discovered



Unseen

Queue: A → C → F → G

Visited: B, D, E, J



BFS Traversal

Step 10: Process node A

Legend

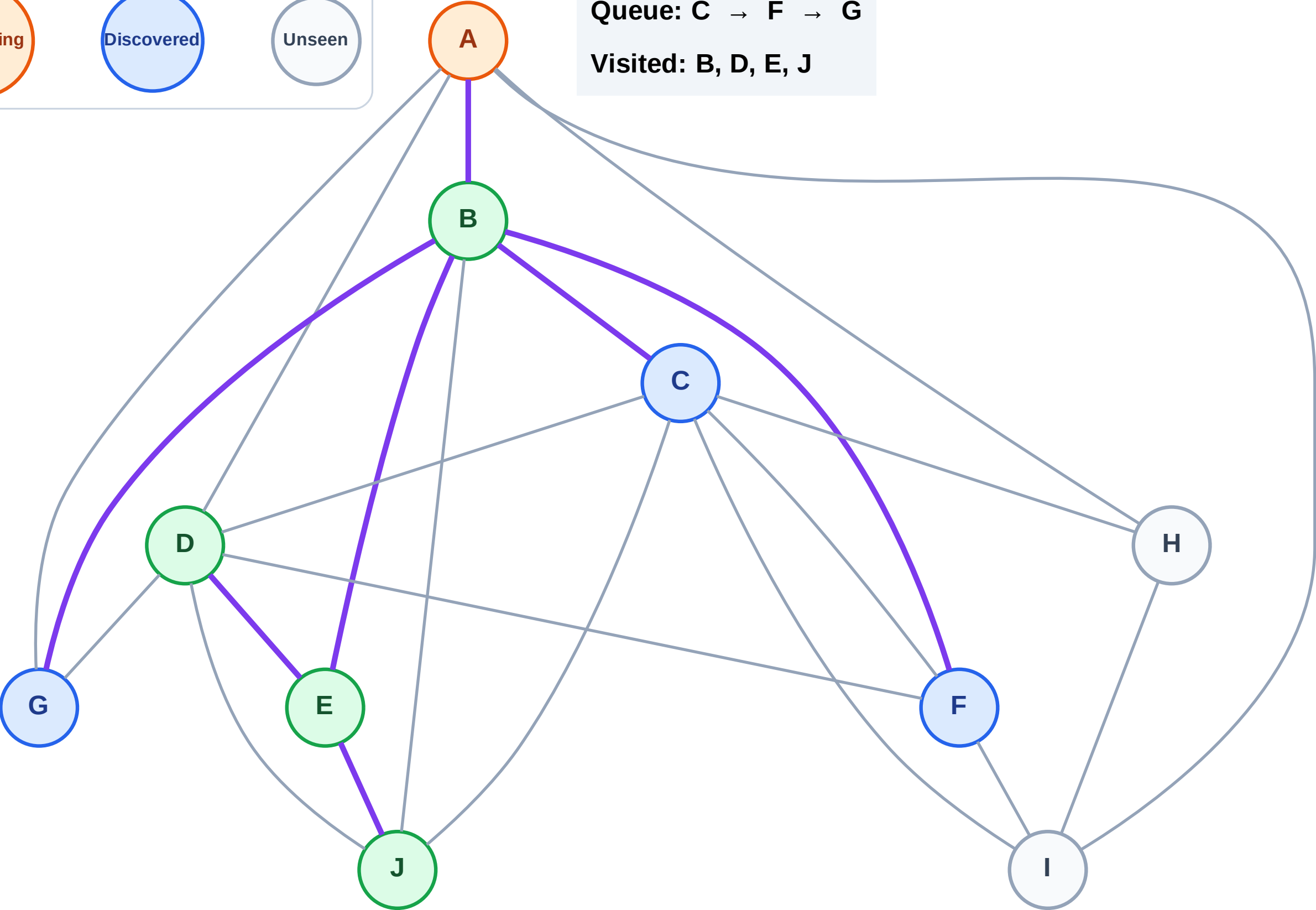
Complete

Processing

Discovered

Unseen

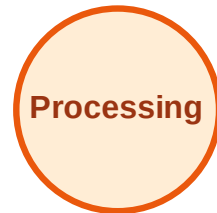
Queue: C → F → G
Visited: B, D, E, J



BFS Traversal

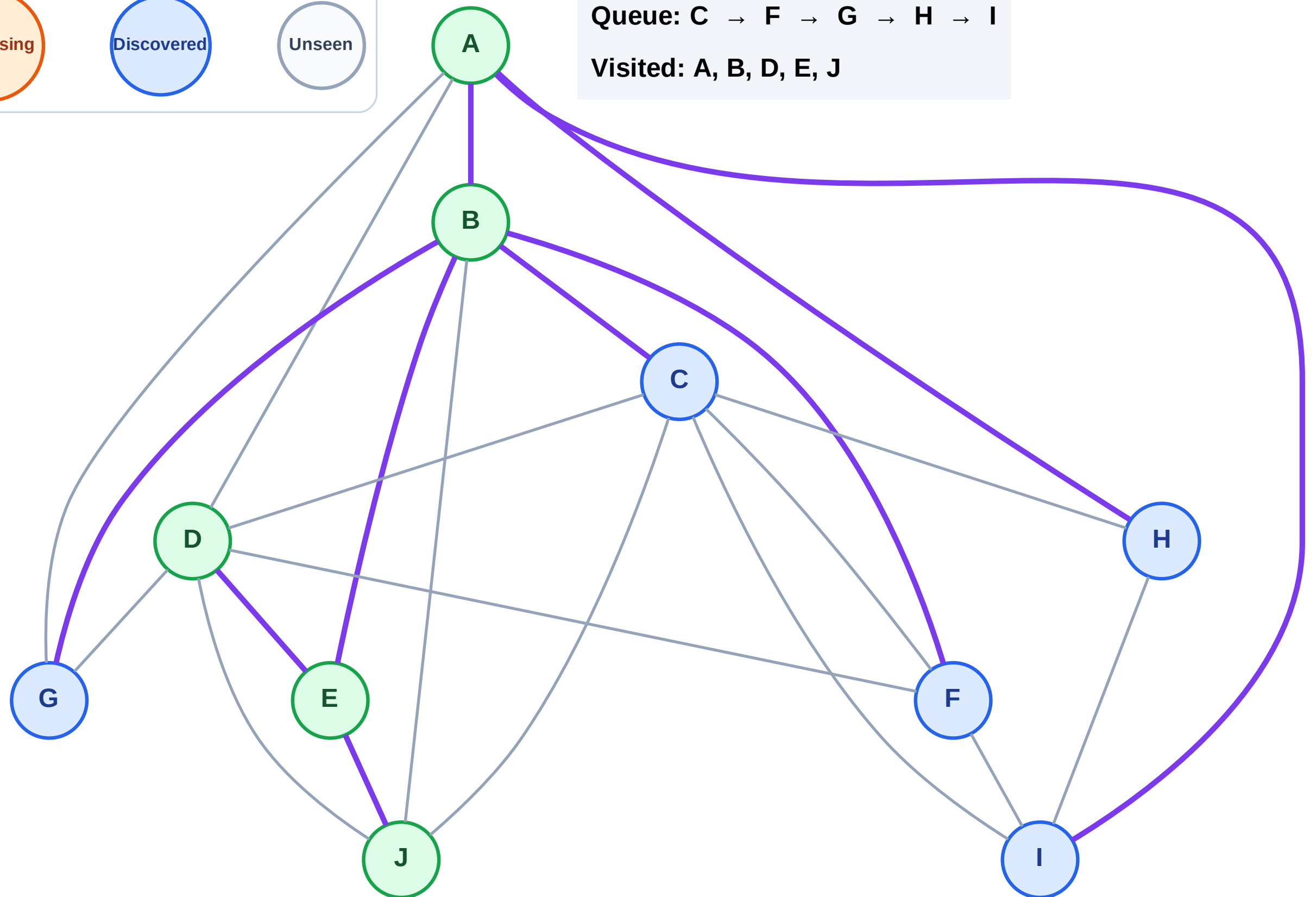
Step 11: Discover H, I from A

Legend



Queue: C → F → G → H → I

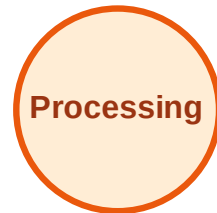
Visited: A, B, D, E, J



BFS Traversal

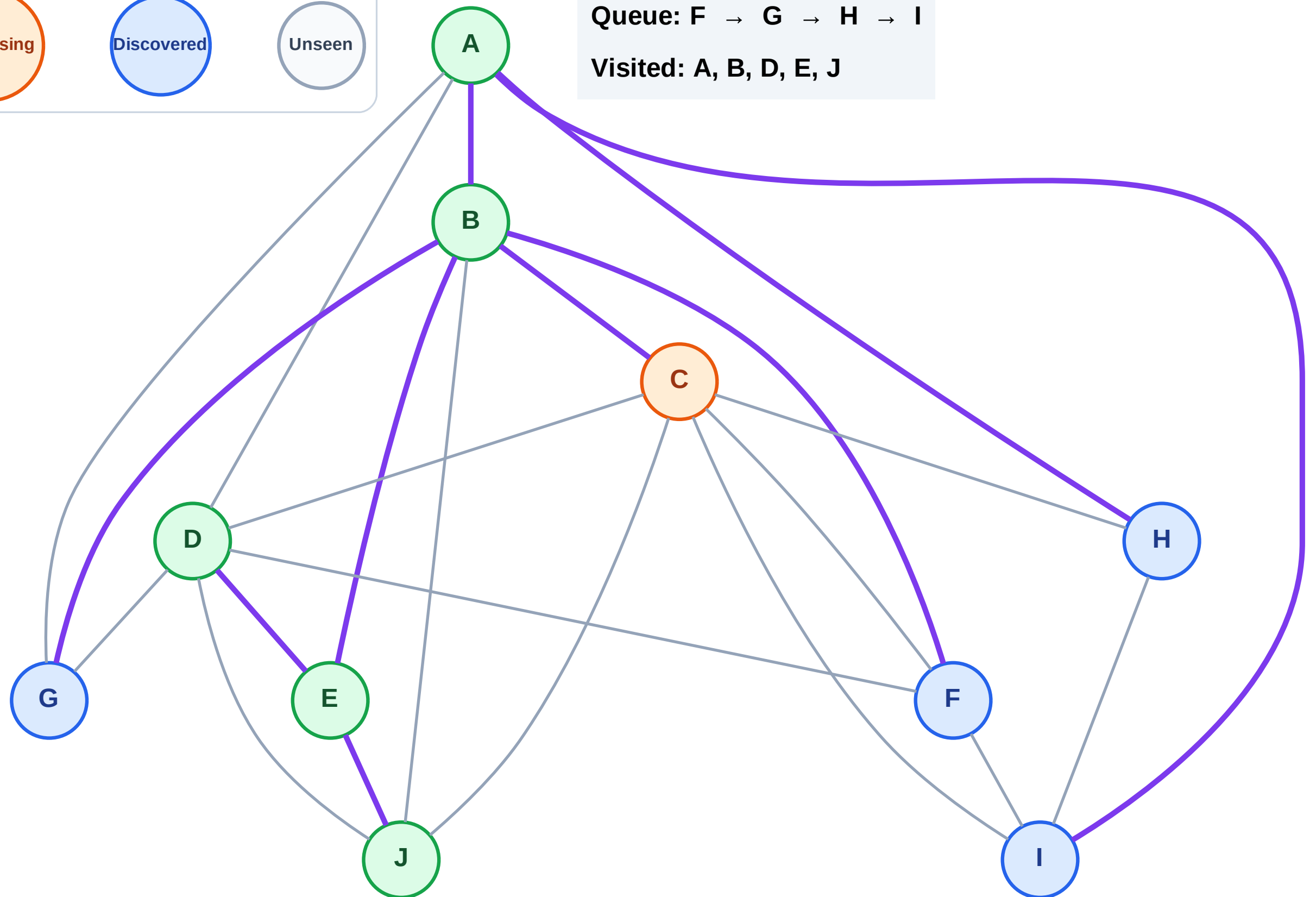
Step 12: Process node C

Legend



Queue: F → G → H → I

Visited: A, B, D, E, J



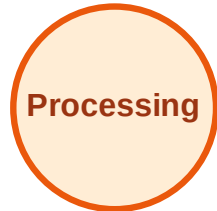
BFS Traversal

Step 13: No new neighbors from C

Legend



Complete



Processing



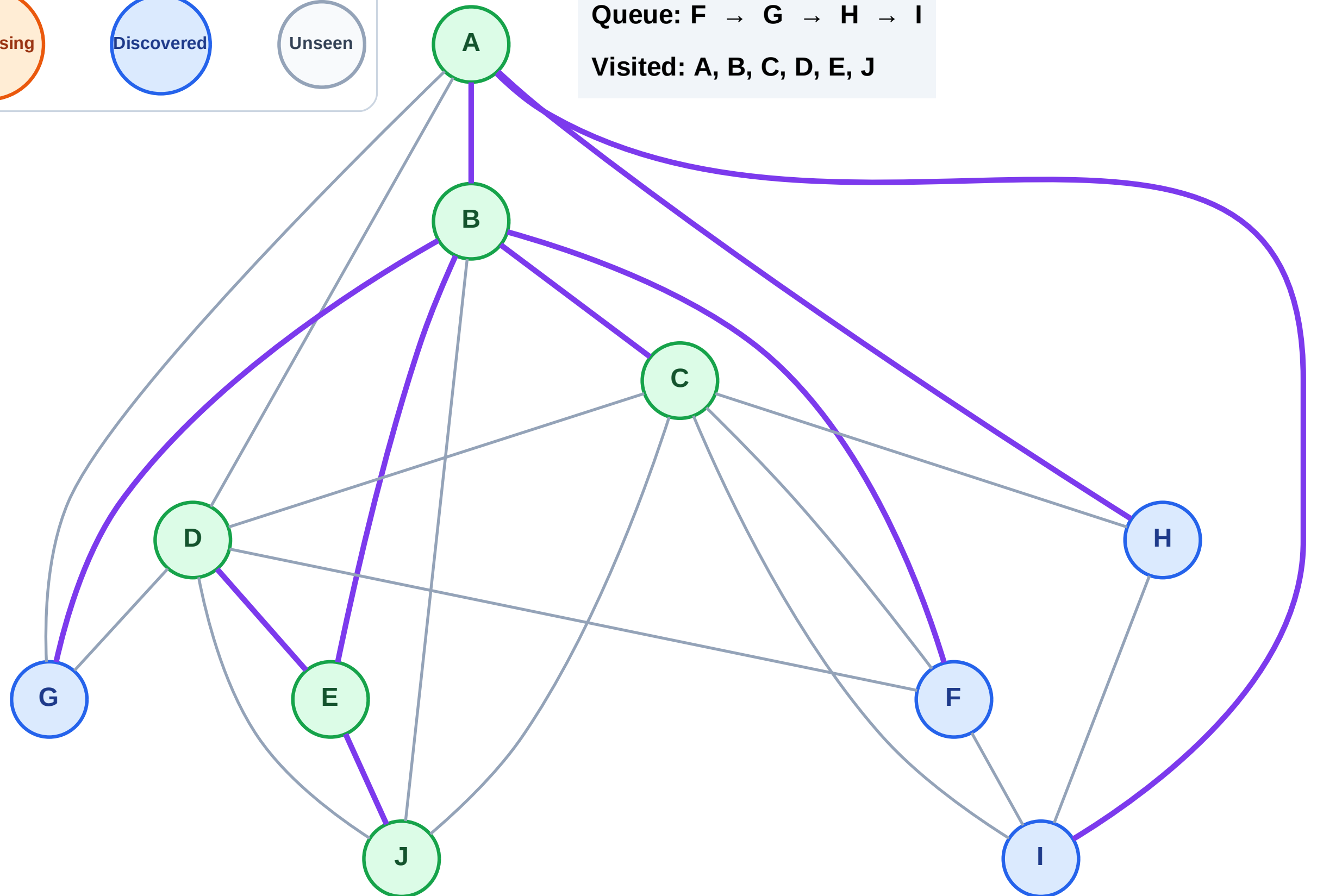
Discovered



Unseen

Queue: F → G → H → I

Visited: A, B, C, D, E, J



BFS Traversal

Step 14: Process node F

Legend

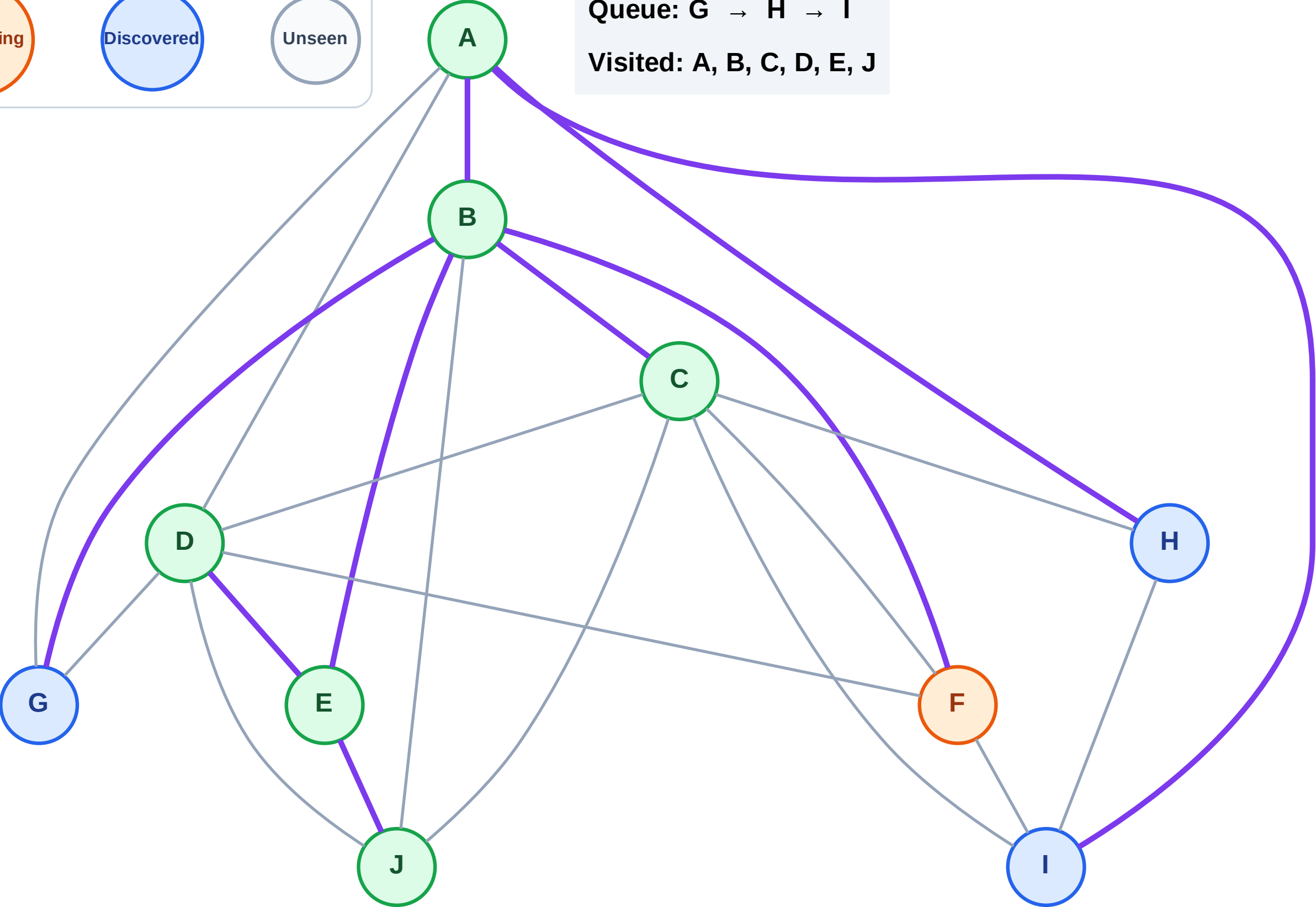
Complete

Processing

Discovered

Unseen

Queue: G → H → I
Visited: A, B, C, D, E, J



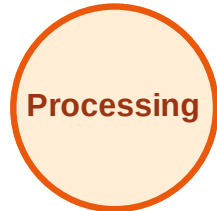
BFS Traversal

Step 15: No new neighbors from F

Legend



Complete



Processing



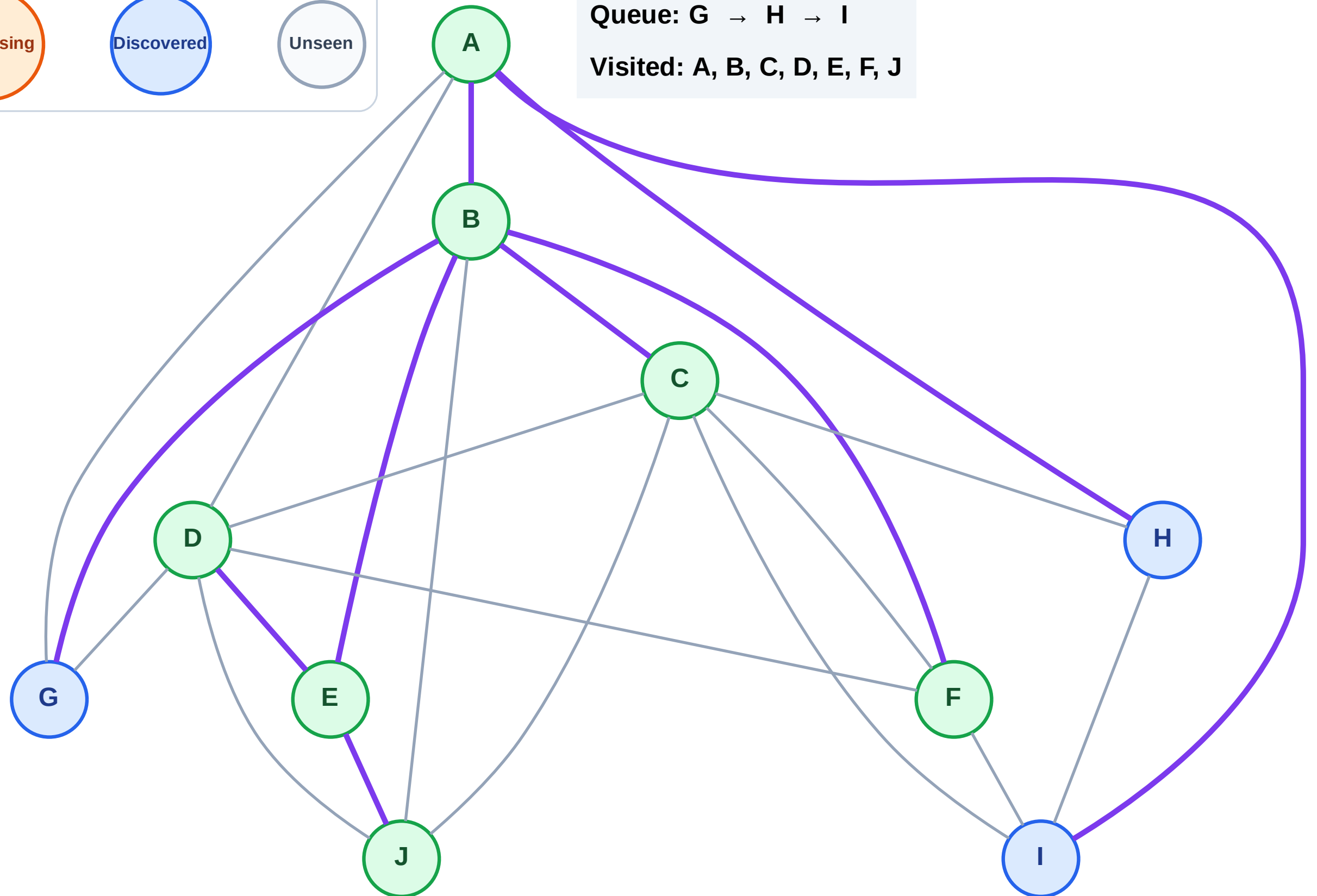
Discovered



Unseen

Queue: G → H → I

Visited: A, B, C, D, E, F, J



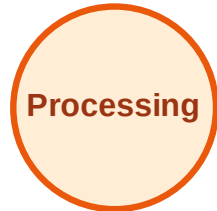
BFS Traversal

Step 16: Process node G

Legend



Complete



Processing



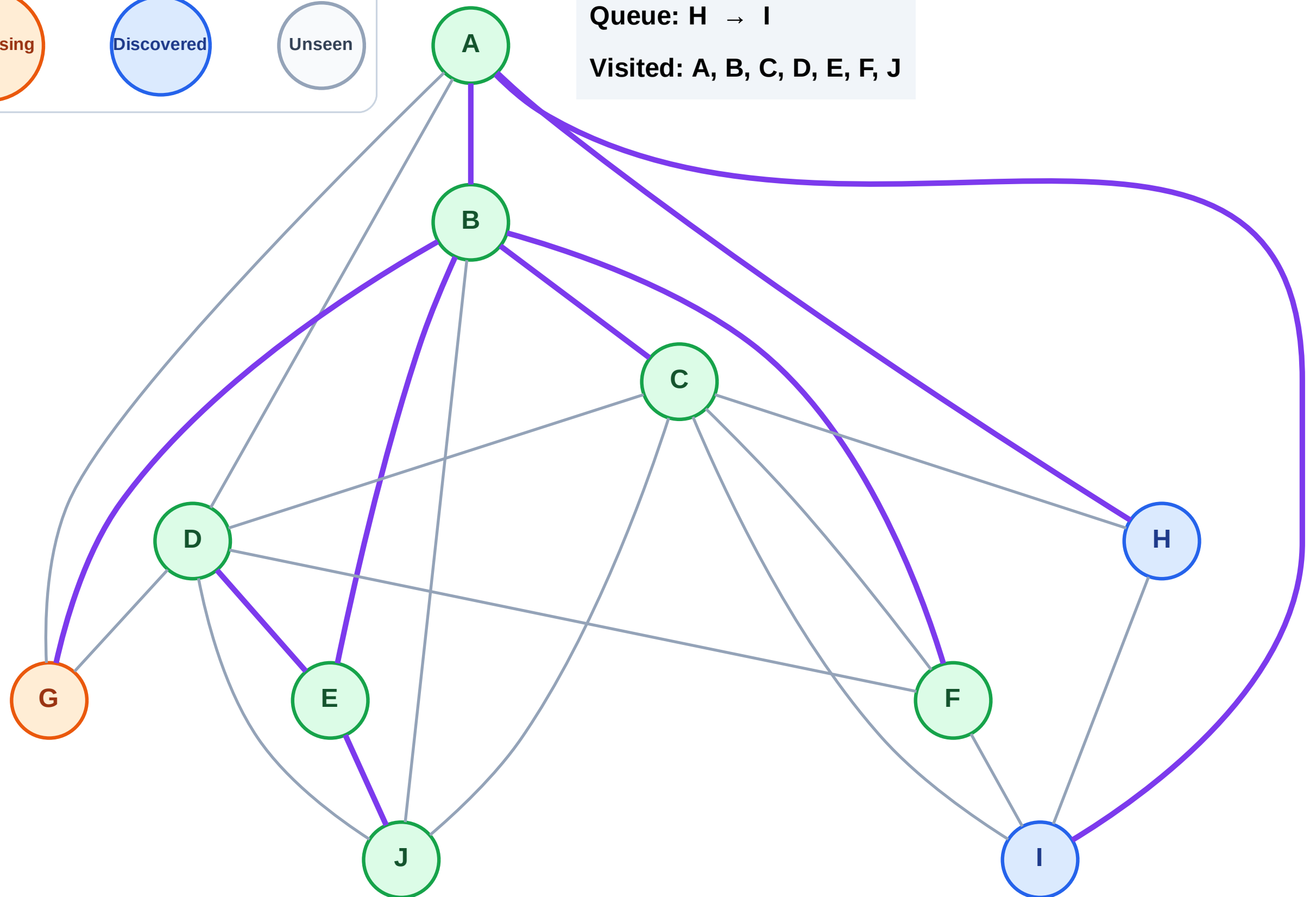
Discovered



Unseen

Queue: H → I

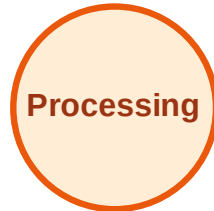
Visited: A, B, C, D, E, F, J



BFS Traversal

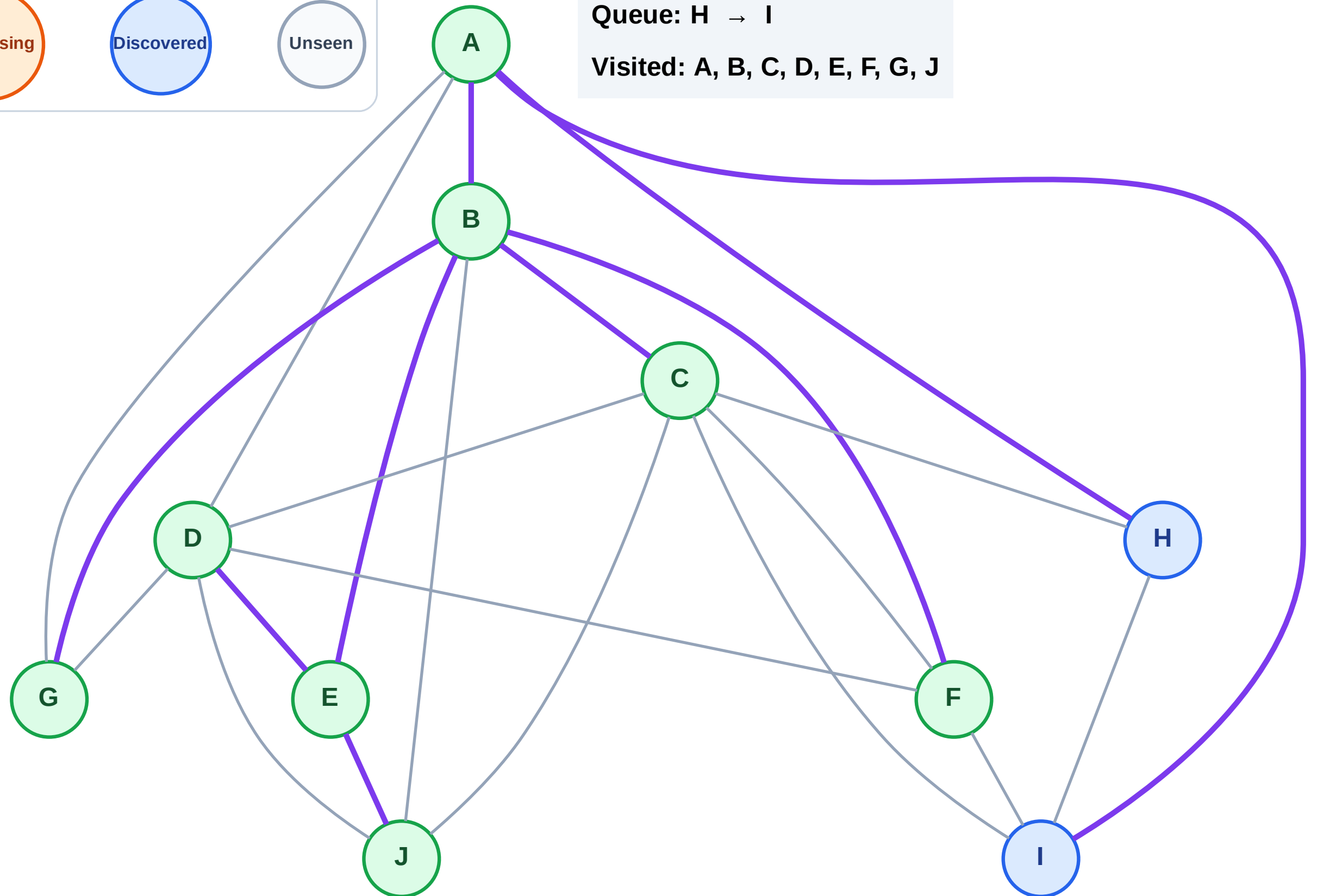
Step 17: No new neighbors from G

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



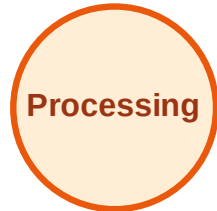
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



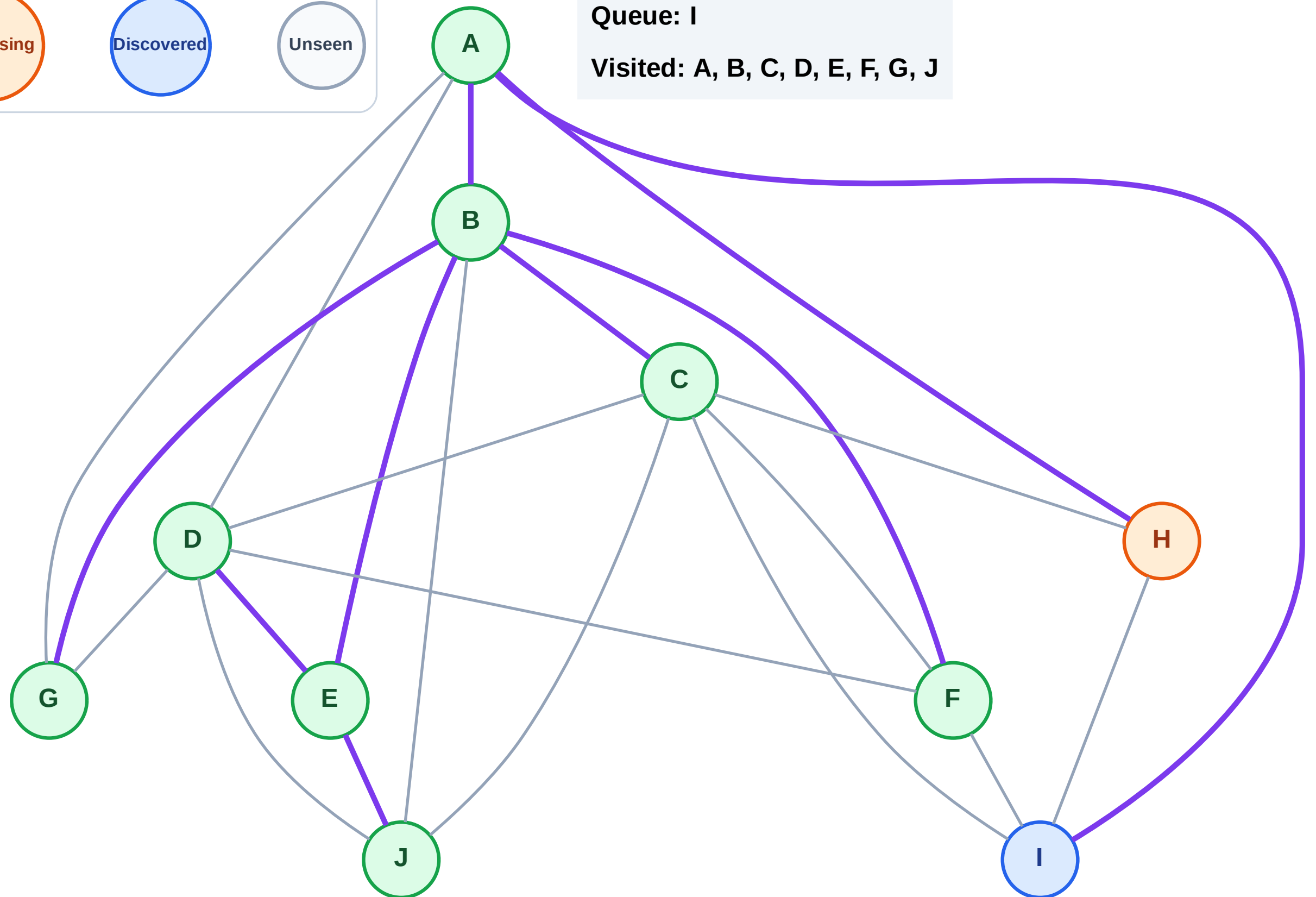
Discovered



Unseen

Queue: I

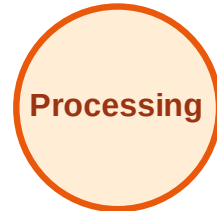
Visited: A, B, C, D, E, F, G, J



BFS Traversal

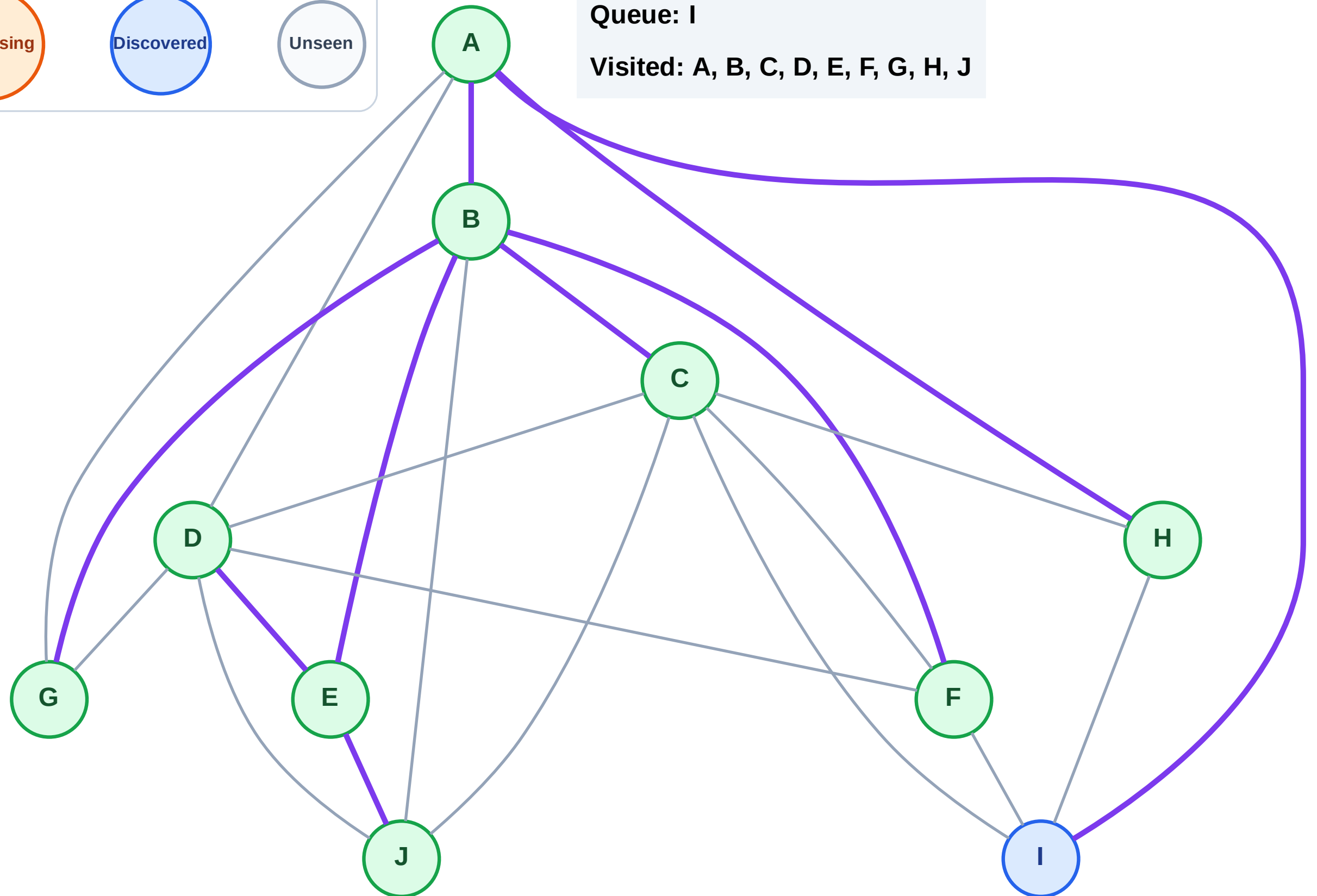
Step 19: No new neighbors from H

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 20: Process node I

Legend

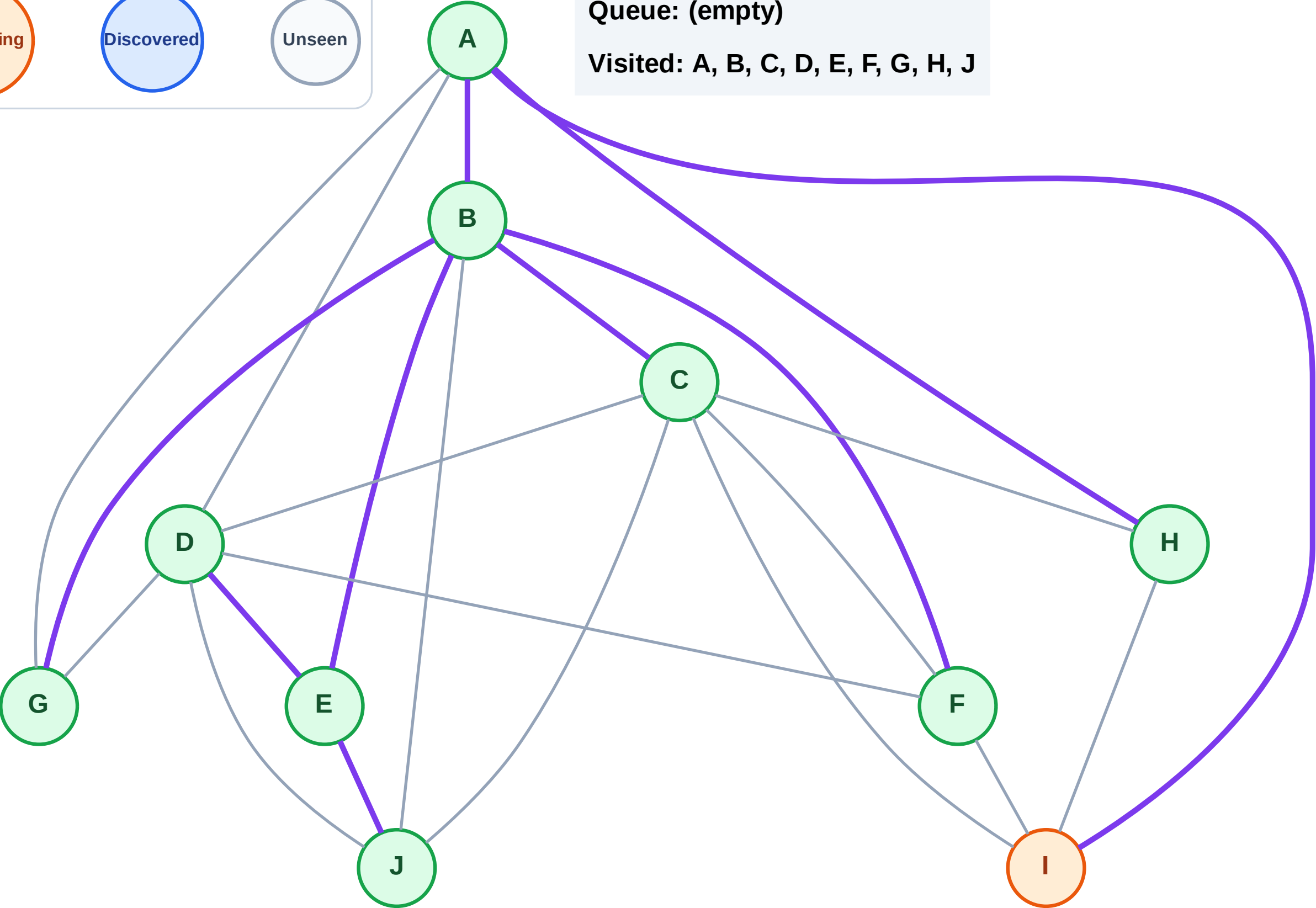
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, J



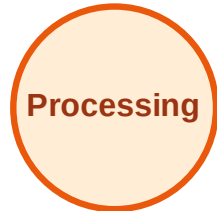
BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



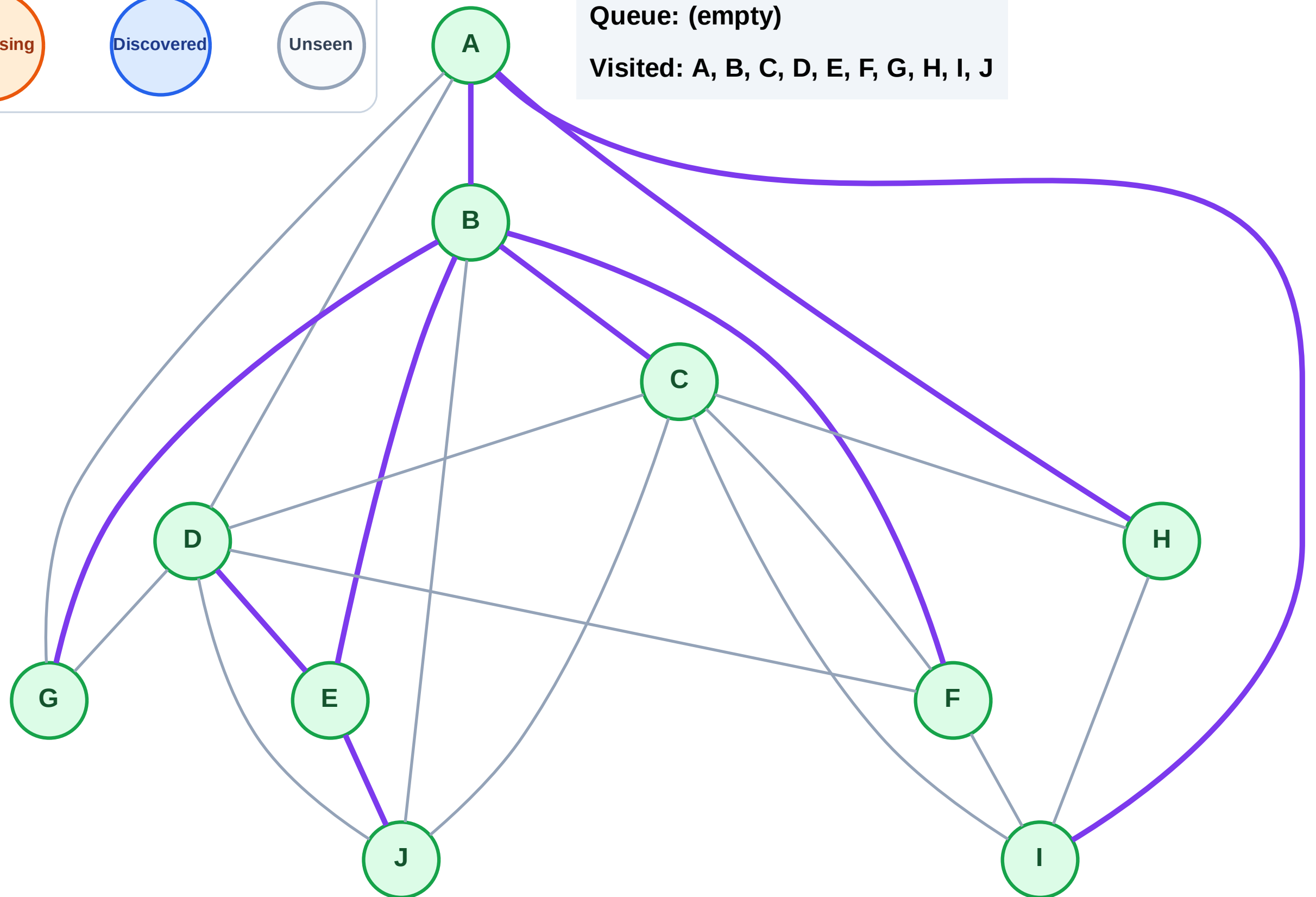
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

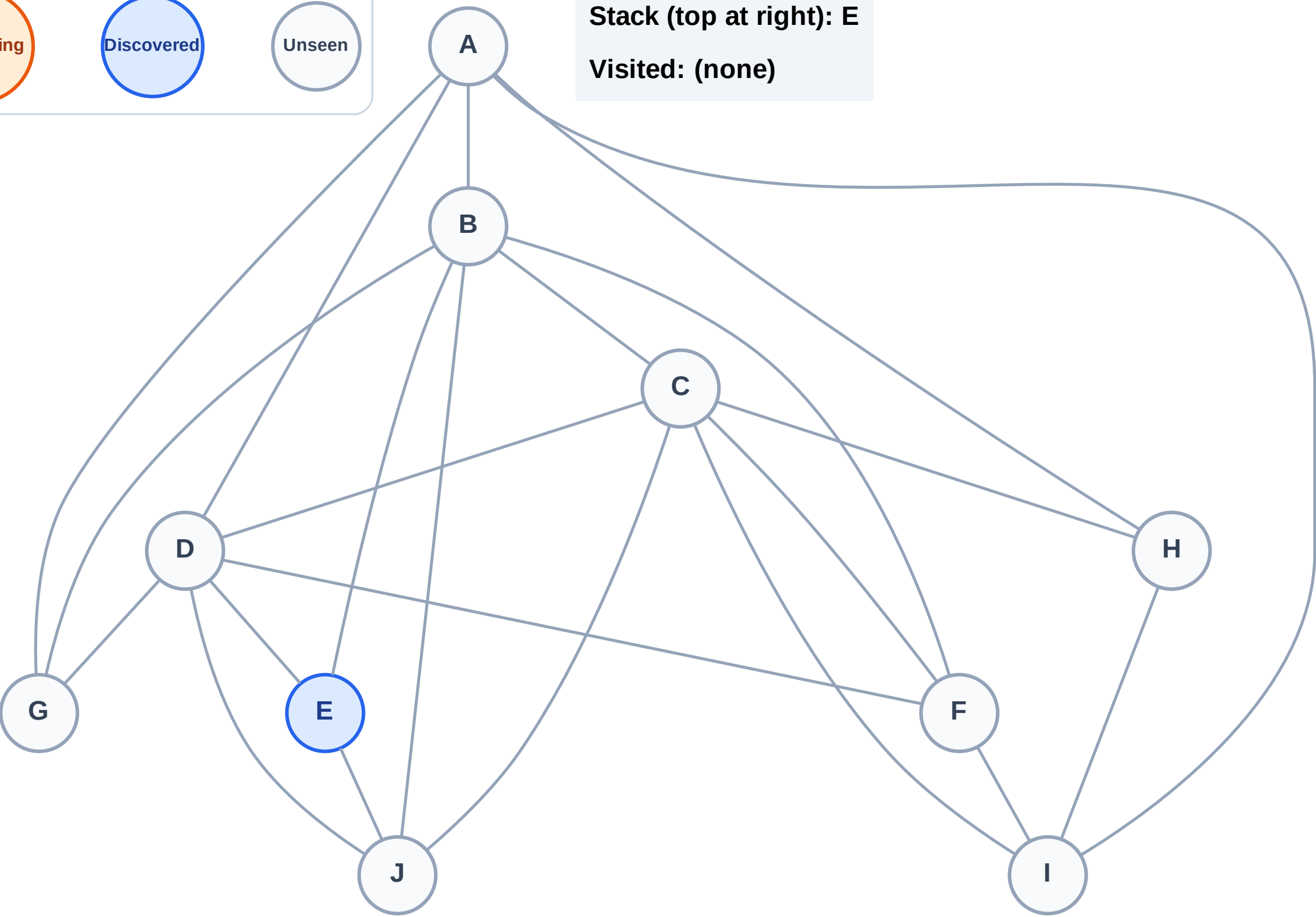
Processing

Discovered

Unseen

Stack (top at right): E

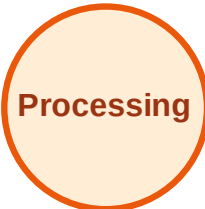
Visited: (none)



DFS Traversal

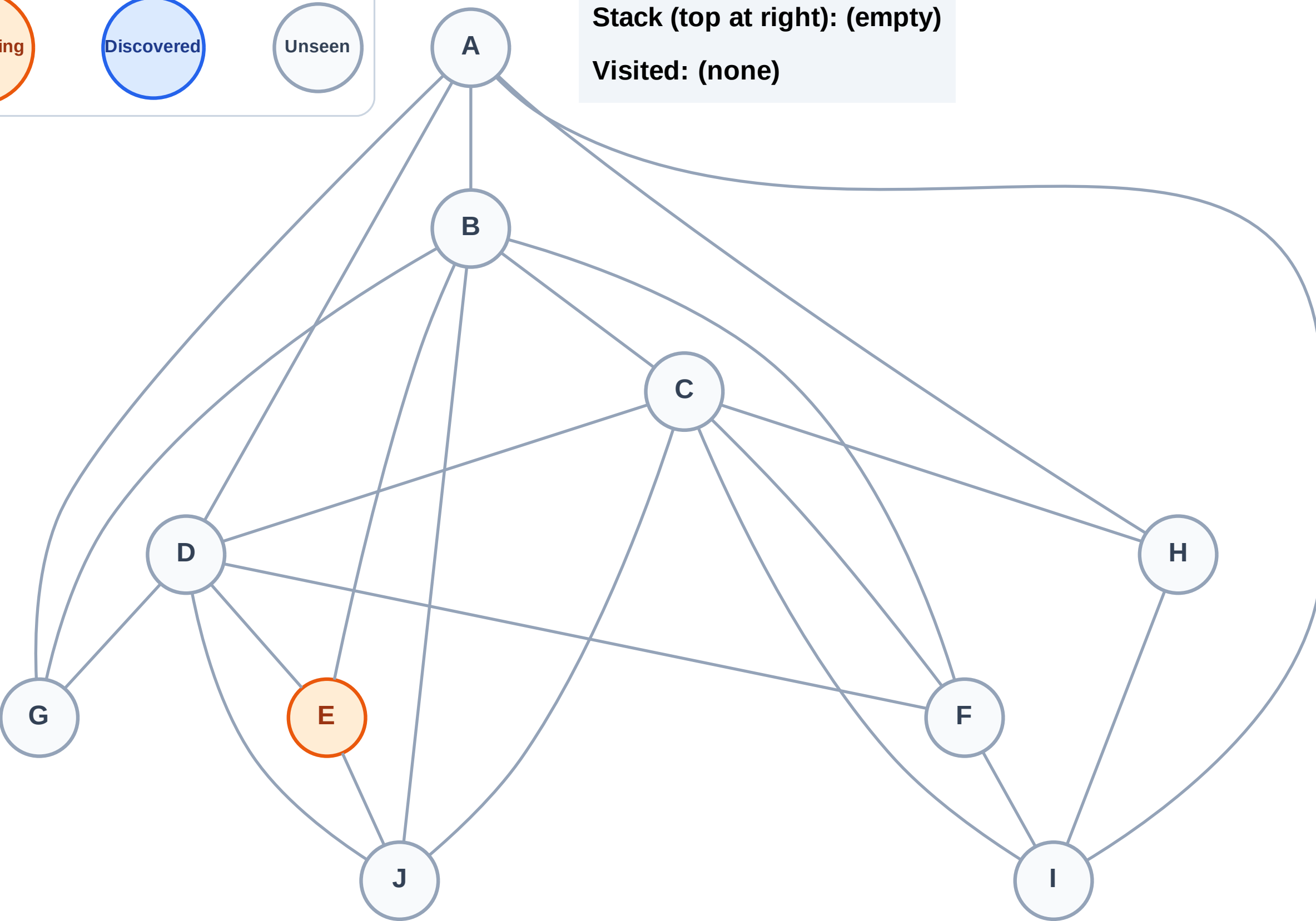
Step 2: Process node E

Legend



Stack (top at right): (empty)

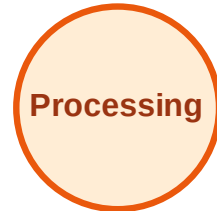
Visited: (none)



DFS Traversal

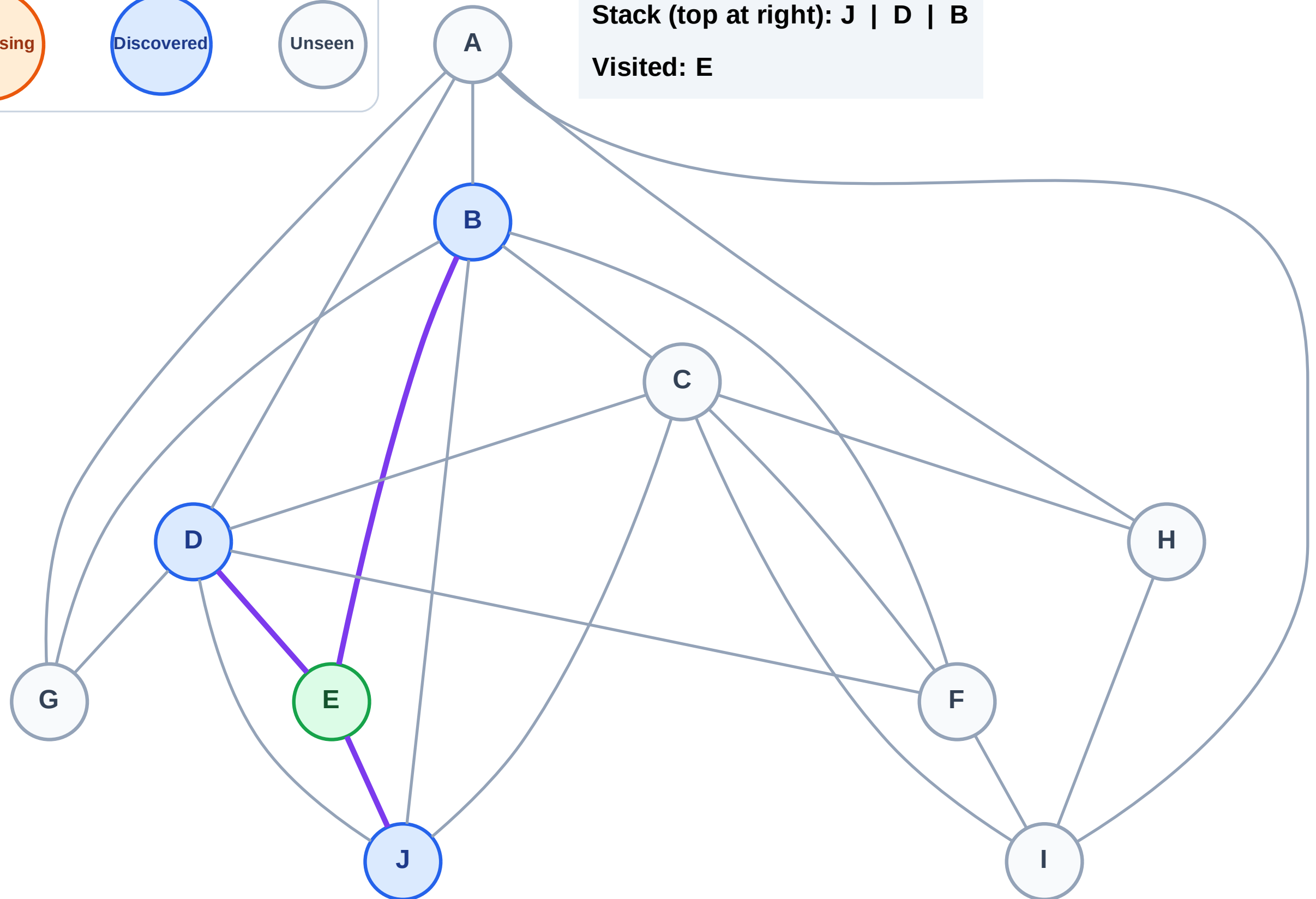
Step 3: Discover J, D, B from E

Legend



Stack (top at right): J | D | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

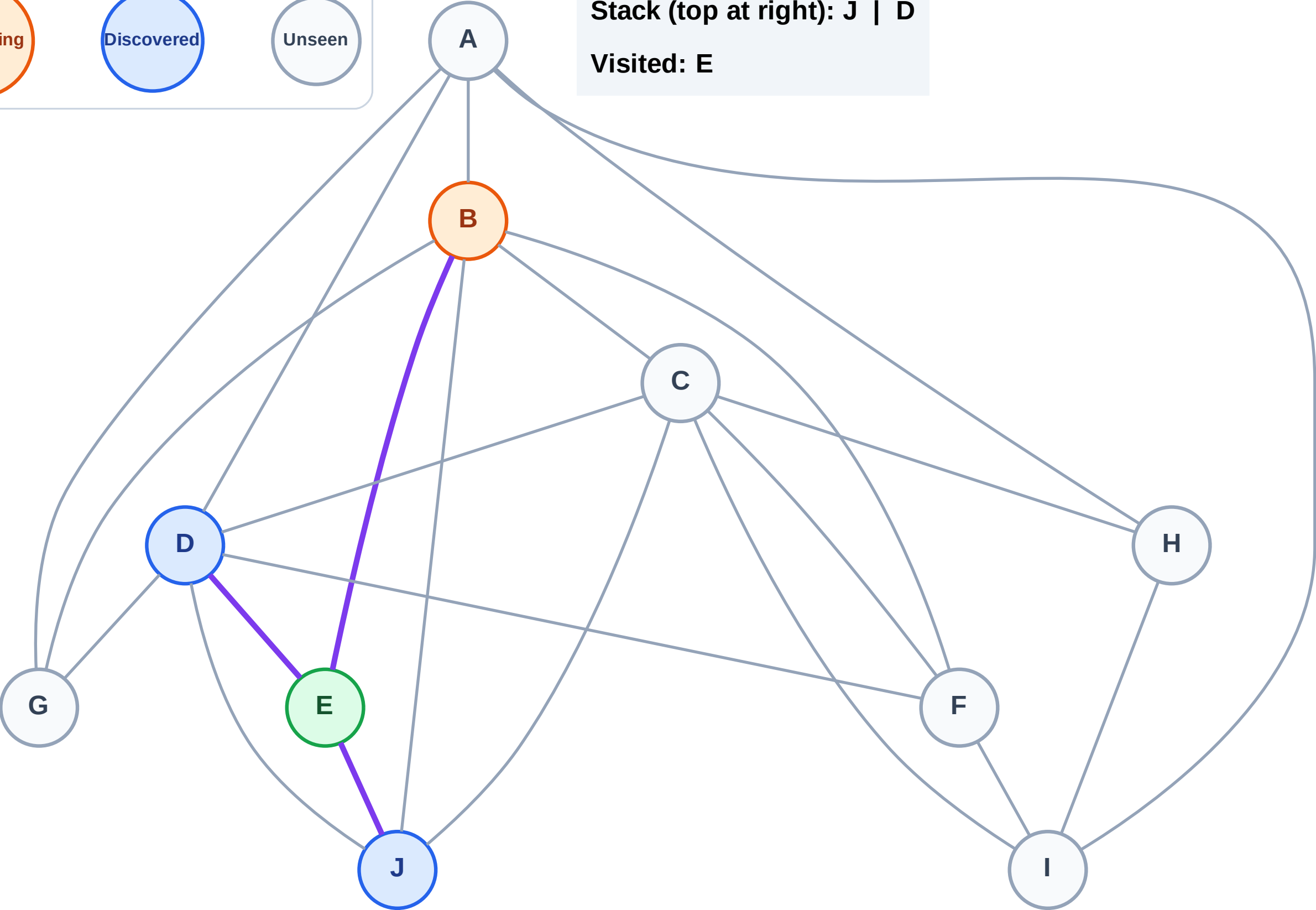
Complete

Processing

Discovered

Unseen

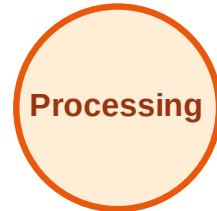
Stack (top at right): J | D
Visited: E



DFS Traversal

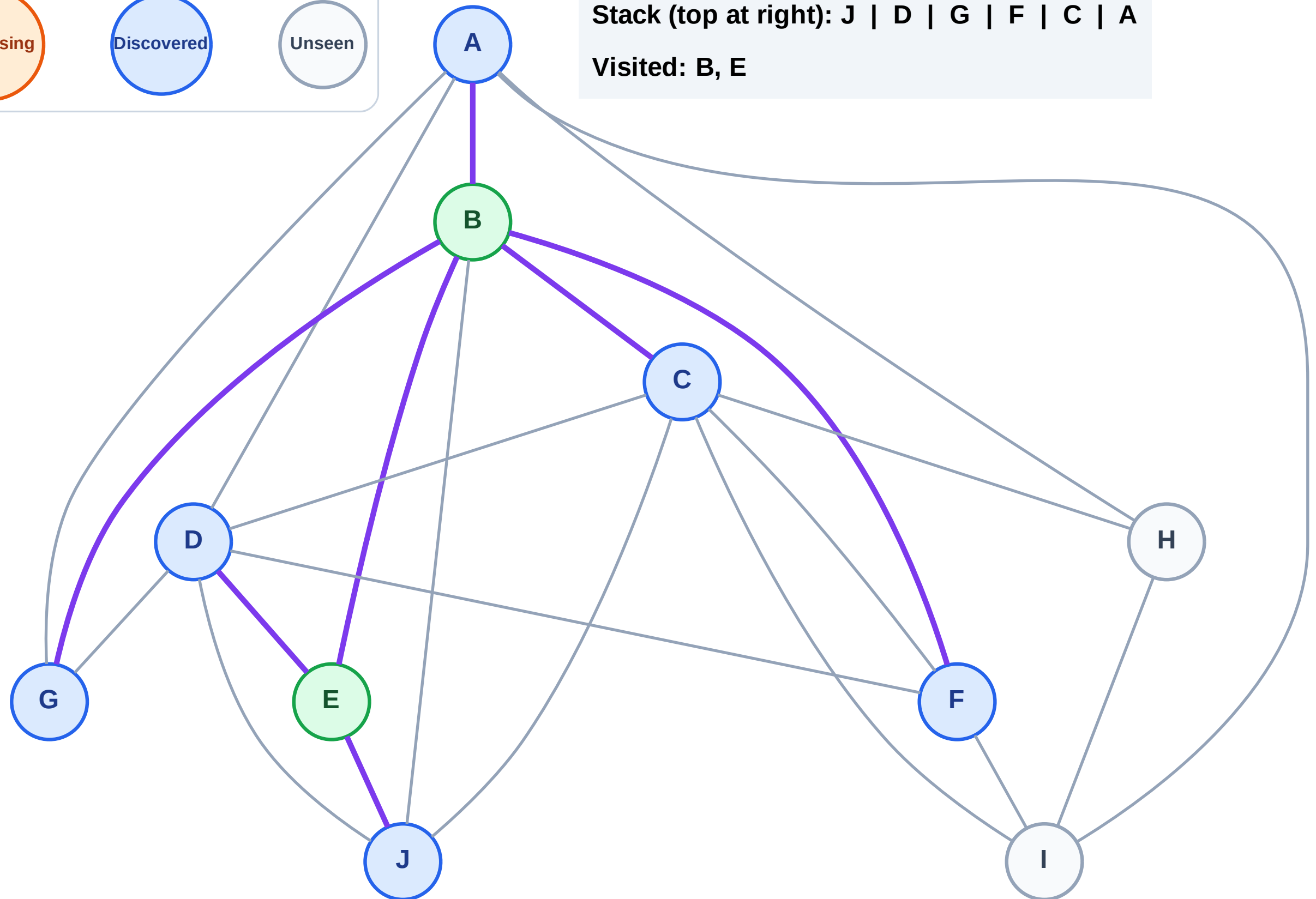
Step 5: Discover G, F, C, A from B

Legend



Stack (top at right): J | D | G | F | C | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

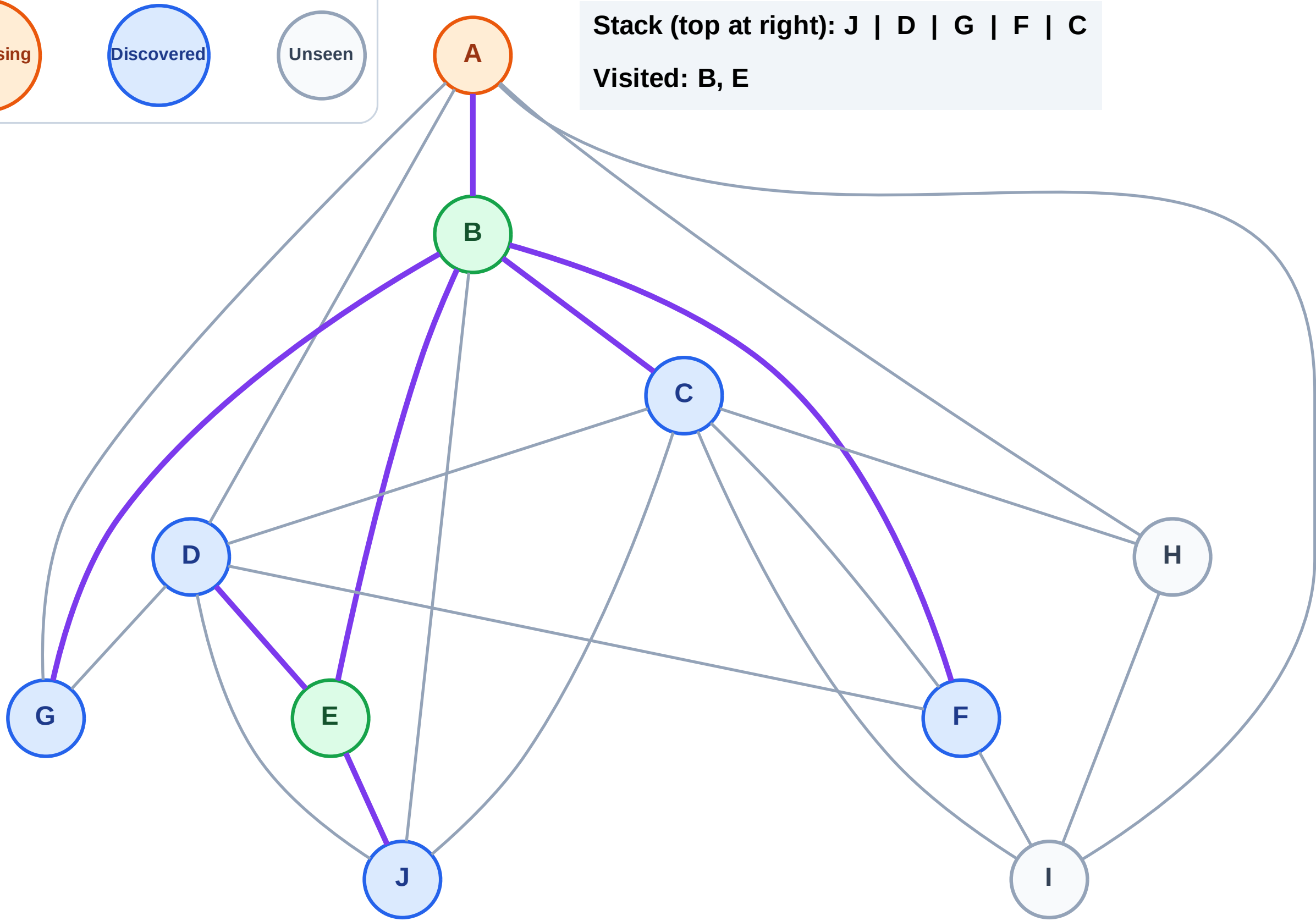
Complete

Processing

Discovered

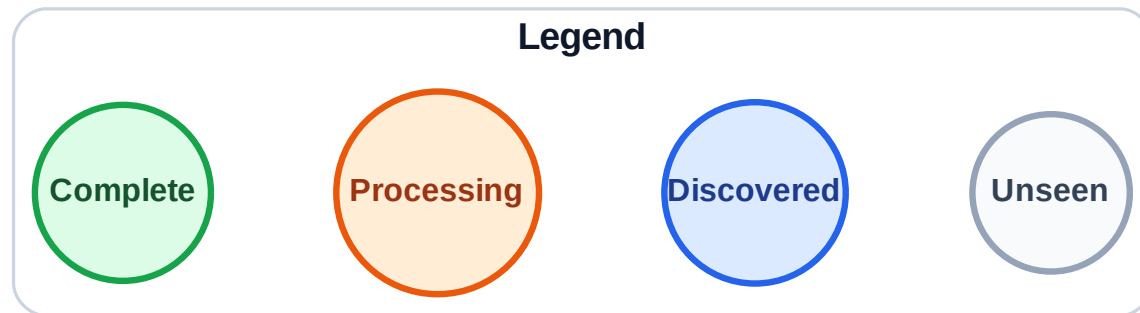
Unseen

Stack (top at right): J | D | G | F | C
Visited: B, E



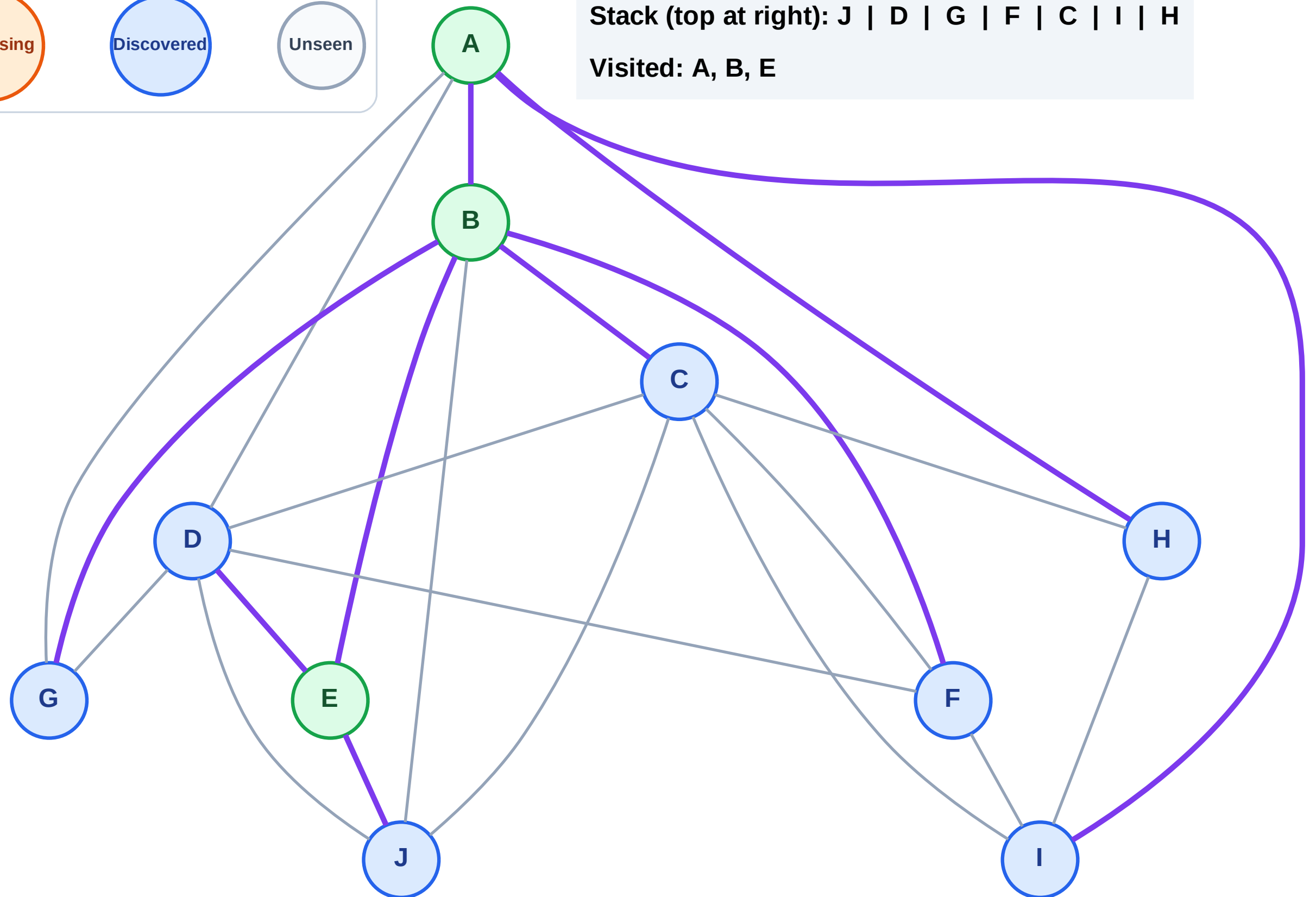
Step 7: Discover I, H from A

Step 7: Discover I, H from A



Stack (top at right): J | D | G | F | C | I | H

Visited: A, B, E



DFS Traversal

Step 8: Process node H

Legend

Complete

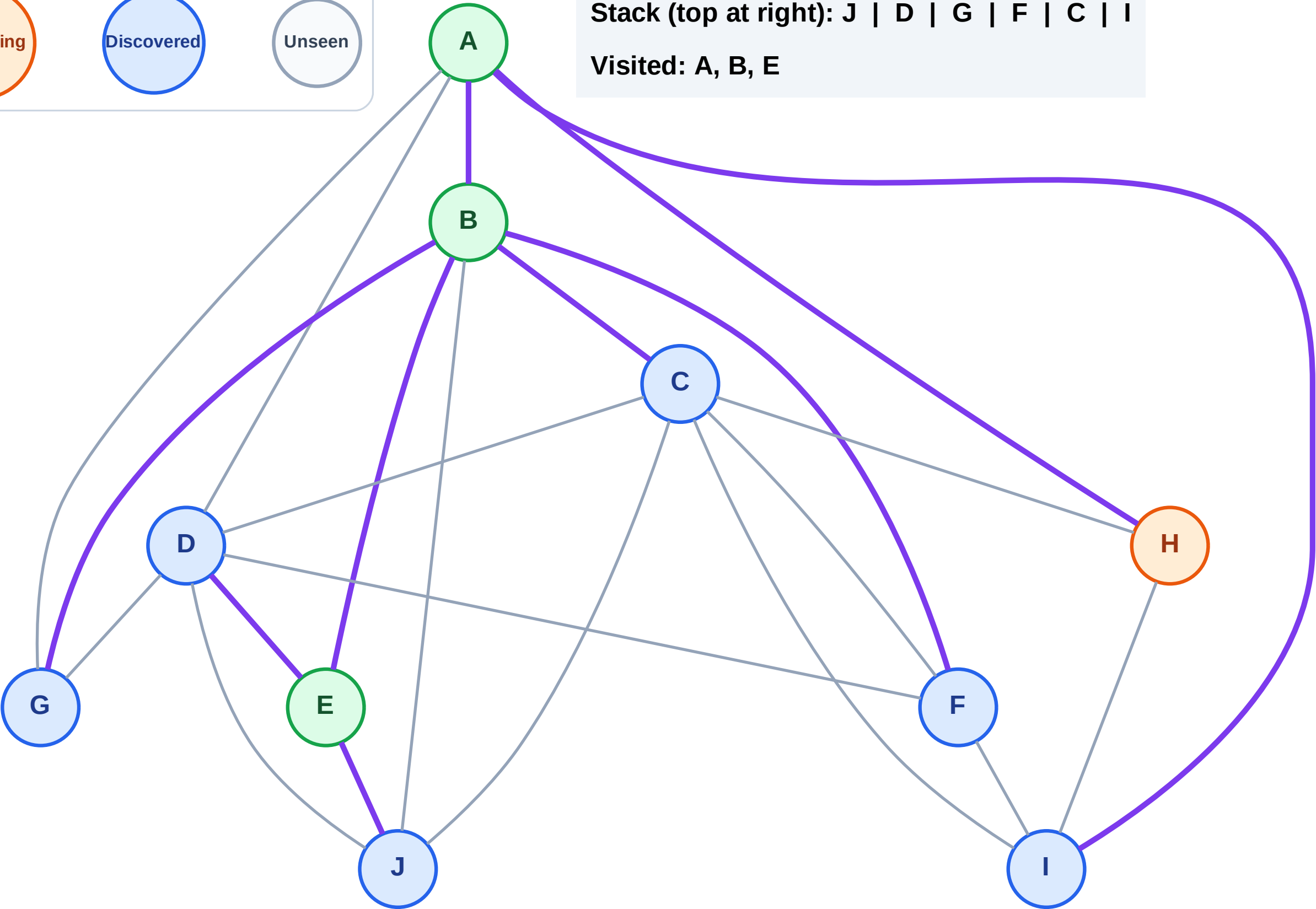
Processing

Discovered

Unseen

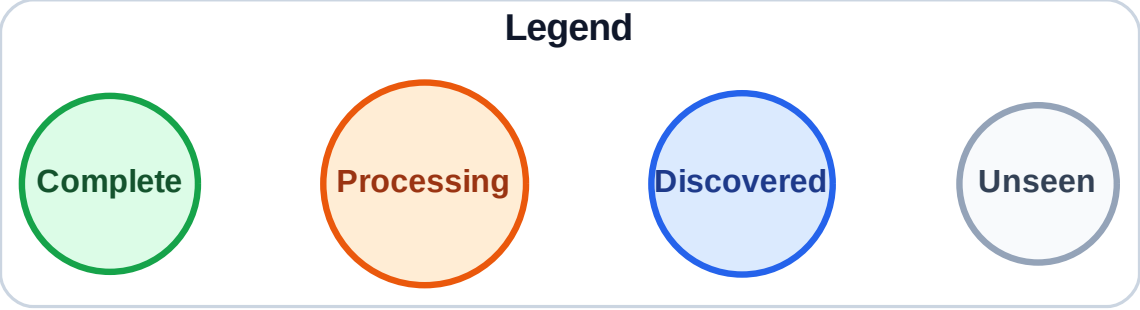
Stack (top at right): J | D | G | F | C | I

Visited: A, B, E



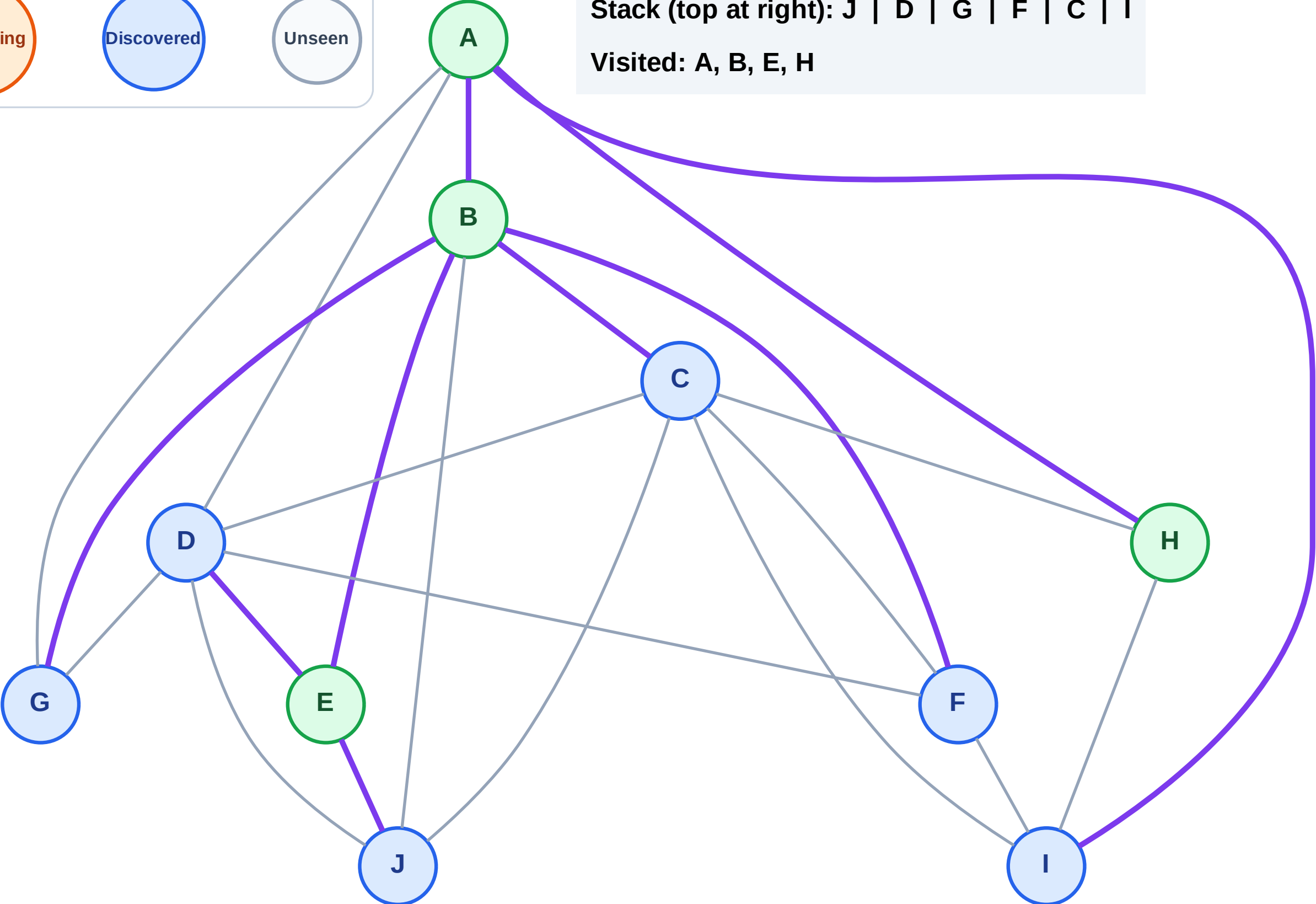
DFS Traversal

Step 9: No new neighbors from H



Stack (top at right): J | D | G | F | C | I

Visited: A, B, E, H



DFS Traversal

Step 10: Process node I

Legend

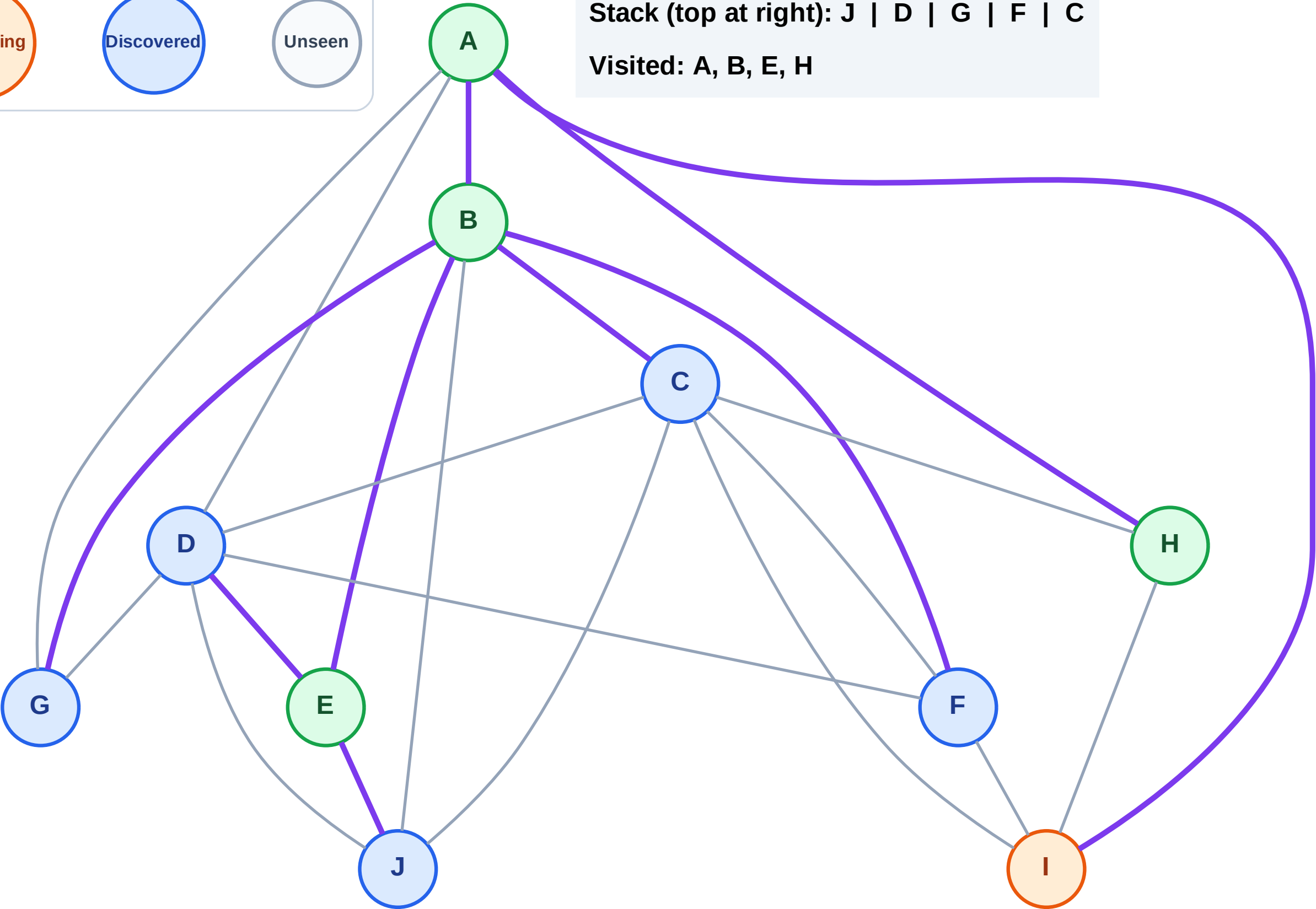
Complete

Processing

Discovered

Unseen

Stack (top at right): J | D | G | F | C
Visited: A, B, E, H



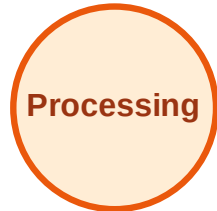
DFS Traversal

Step 11: No new neighbors from I

Legend



Complete



Processing



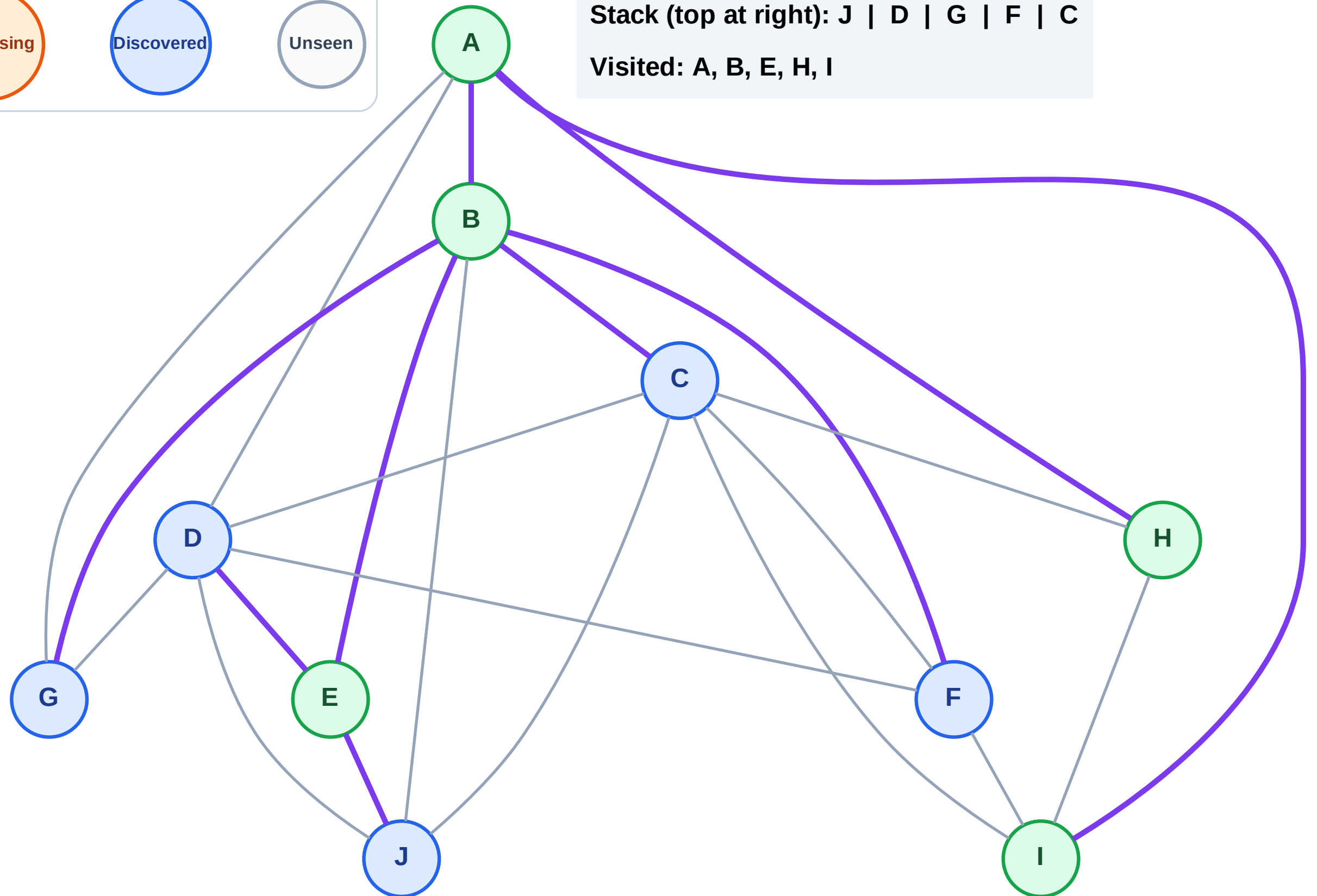
Discovered



Unseen

Stack (top at right): J | D | G | F | C

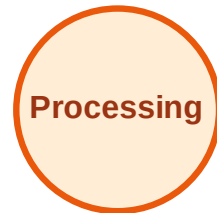
Visited: A, B, E, H, I



DFS Traversal

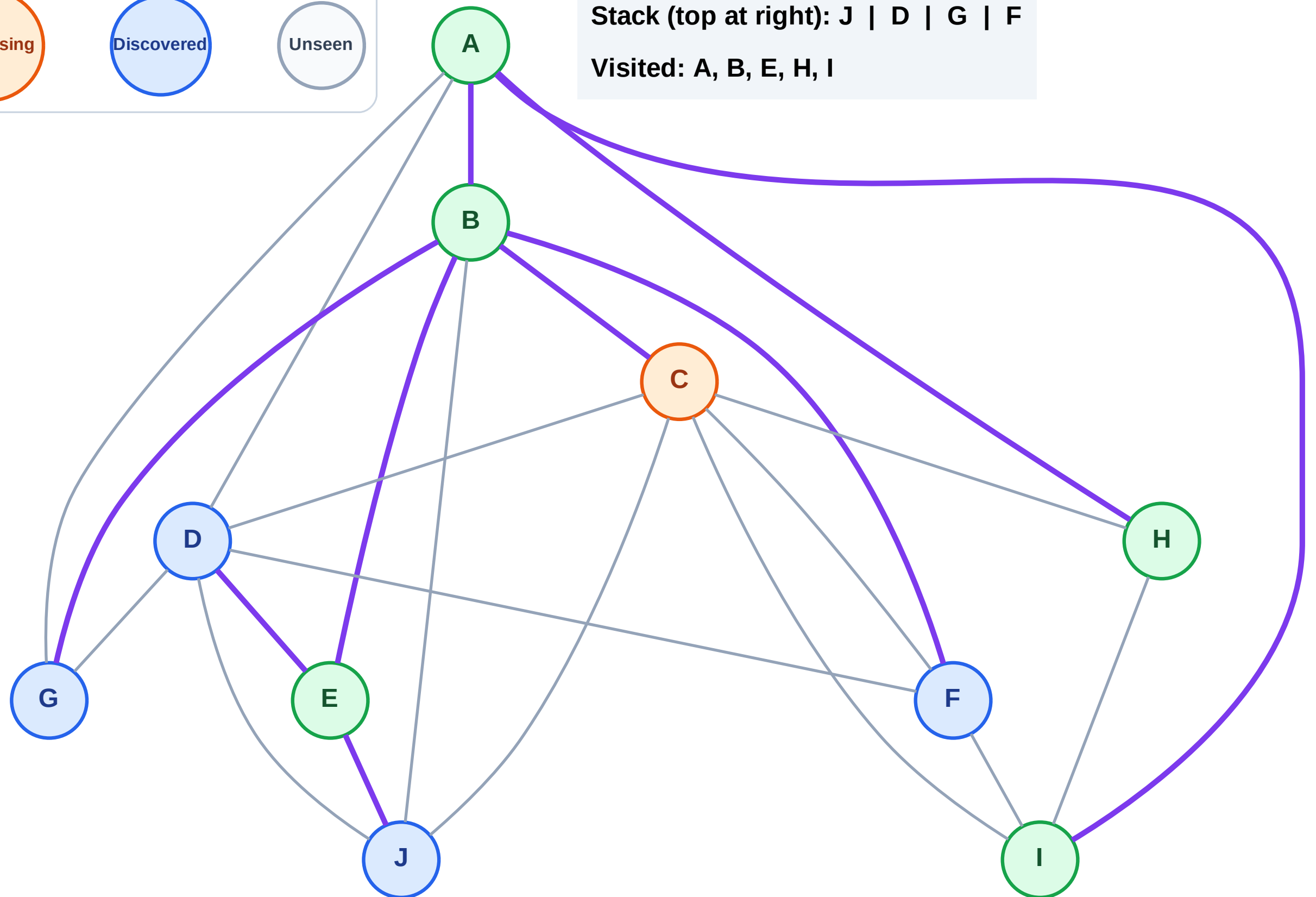
Step 12: Process node C

Legend



Stack (top at right): J | D | G | F

Visited: A, B, E, H, I



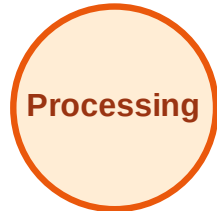
DFS Traversal

Step 13: No new neighbors from C

Legend



Complete



Processing



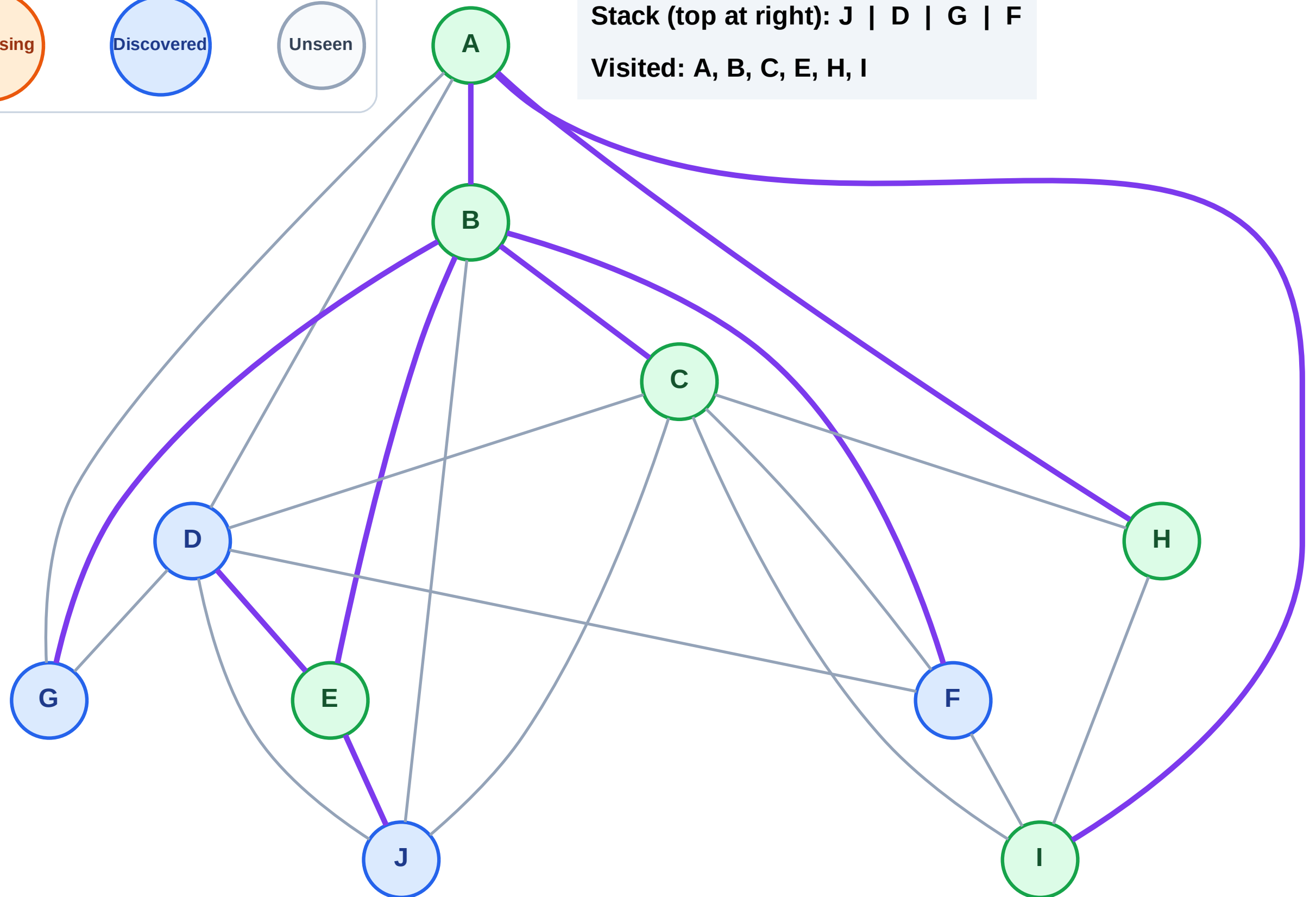
Discovered



Unseen

Stack (top at right): J | D | G | F

Visited: A, B, C, E, H, I



DFS Traversal

Step 14: Process node F

Legend

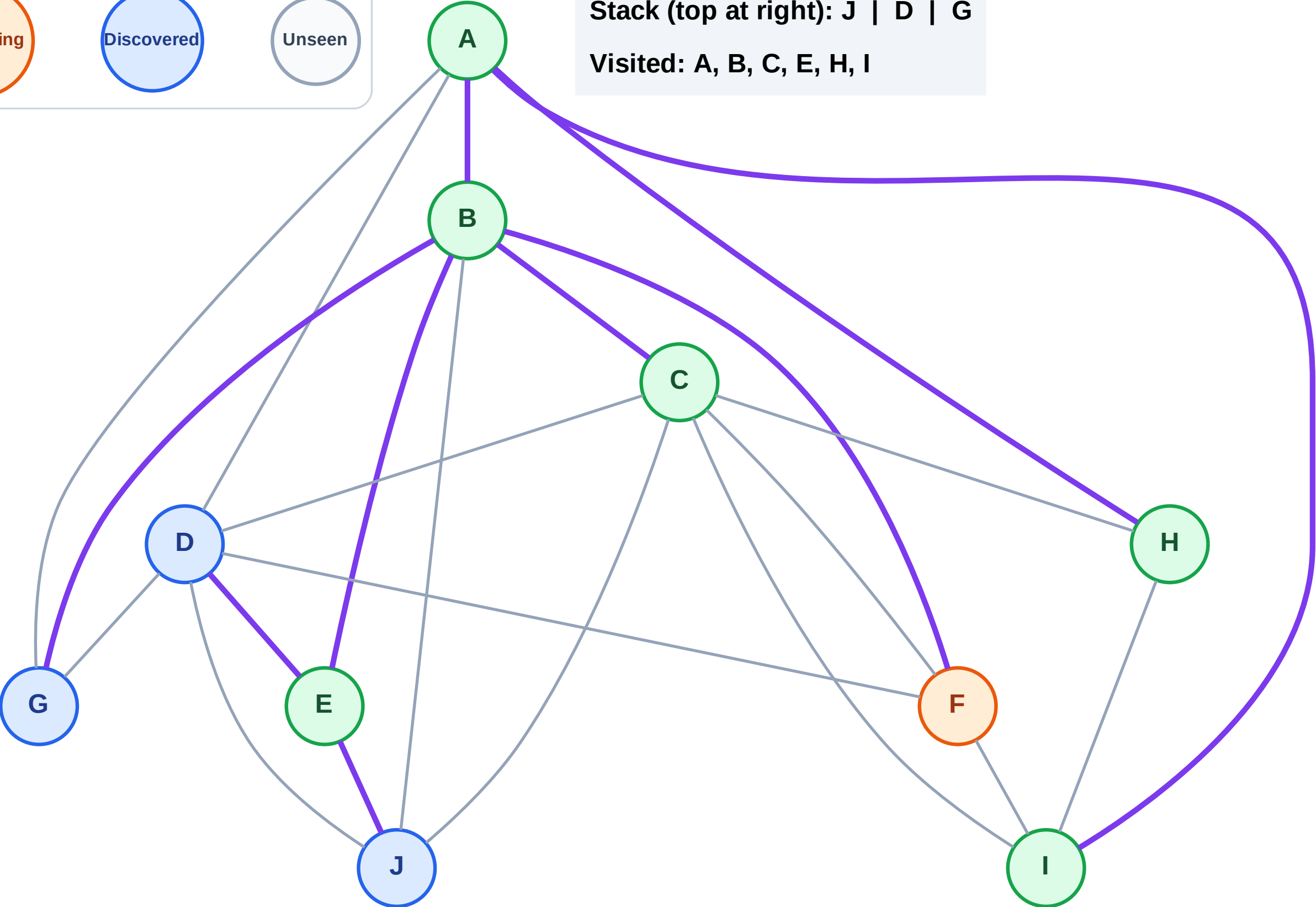
Complete

Processing

Discovered

Unseen

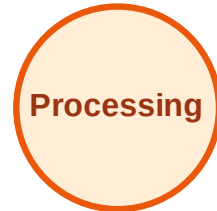
Stack (top at right): J | D | G
Visited: A, B, C, E, H, I



DFS Traversal

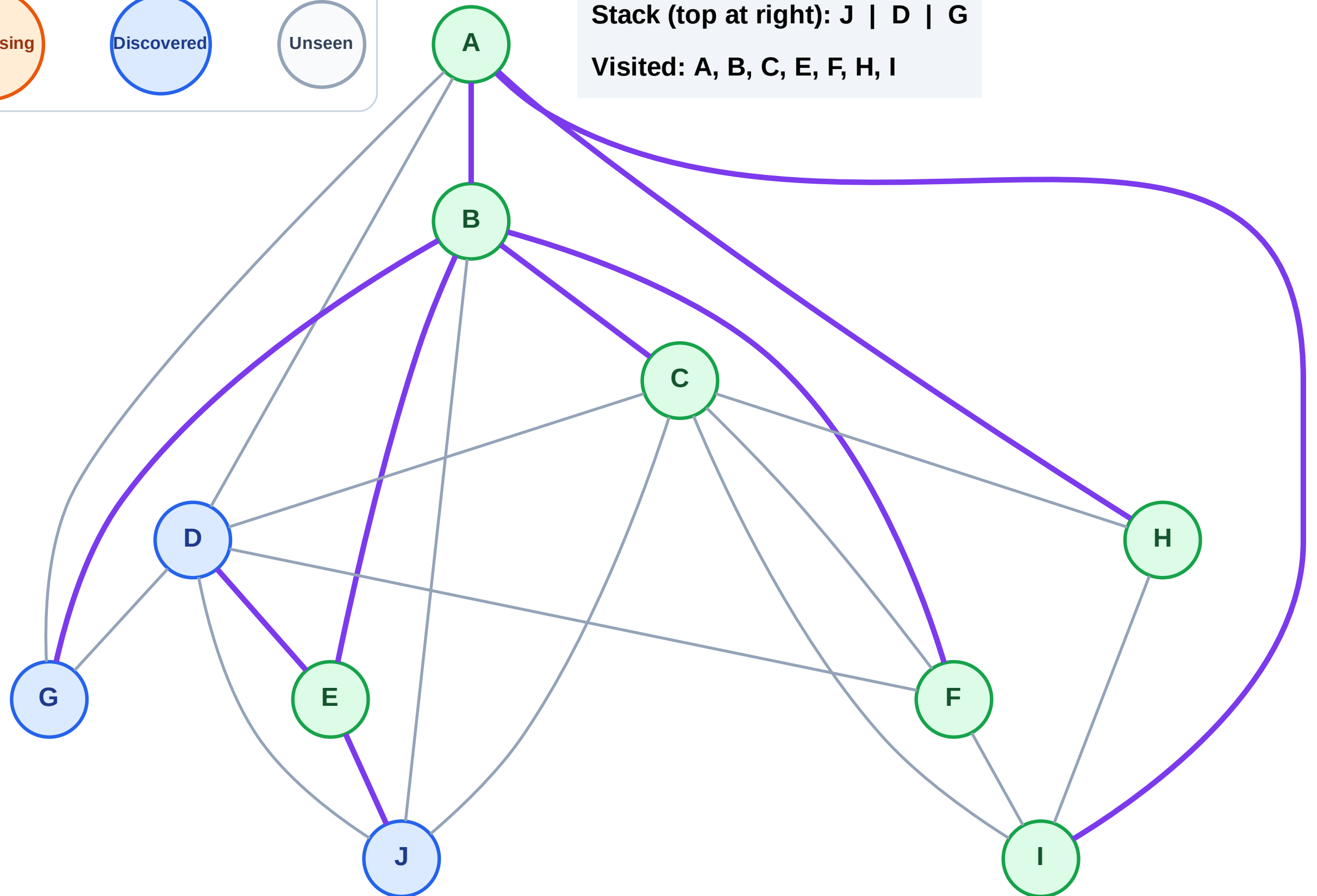
Step 15: No new neighbors from F

Legend



Stack (top at right): J | D | G

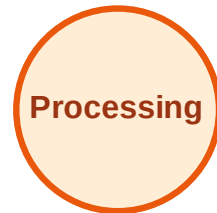
Visited: A, B, C, E, F, H, I



DFS Traversal

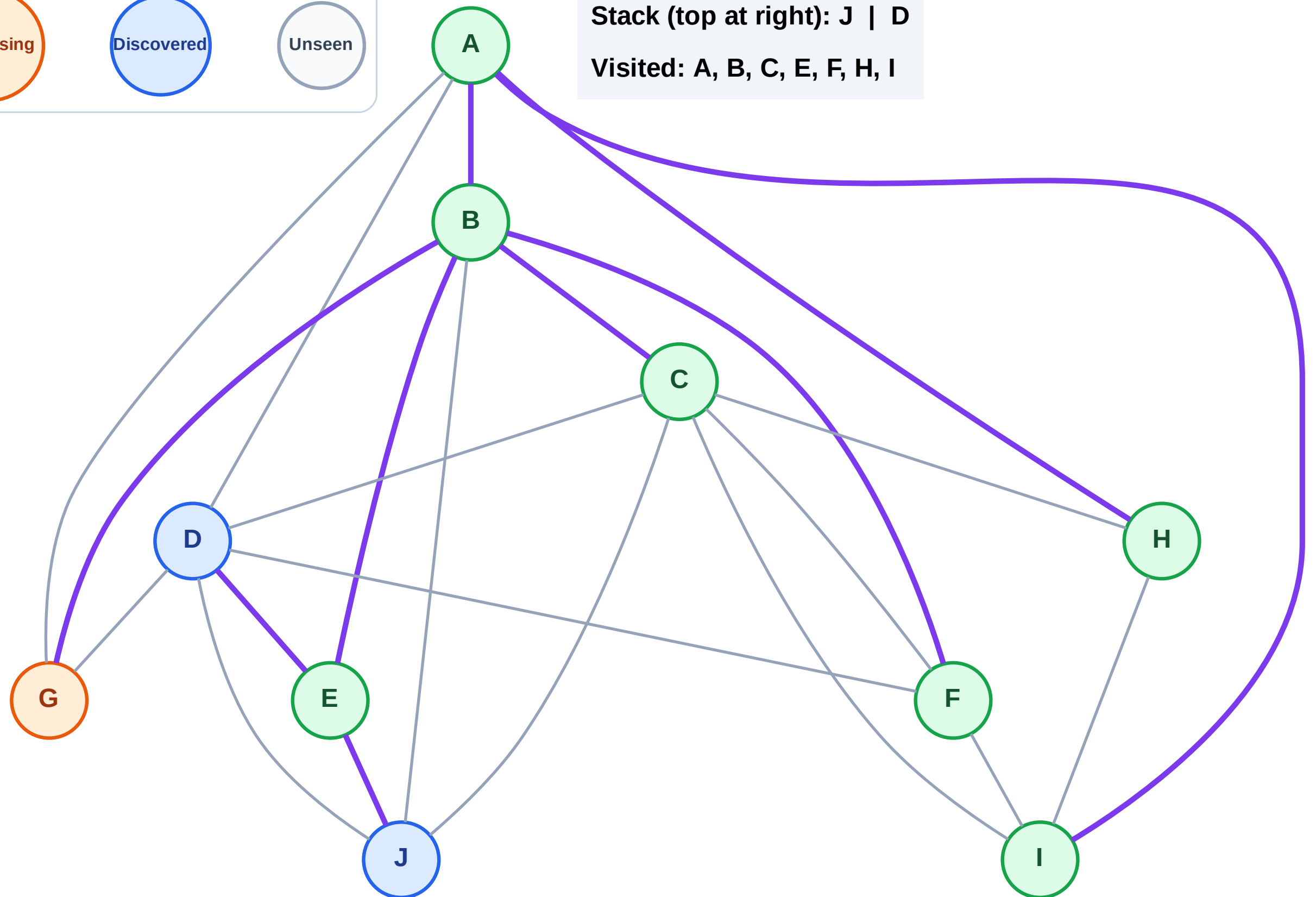
Step 16: Process node G

Legend



Stack (top at right): J | D

Visited: A, B, C, E, F, H, I



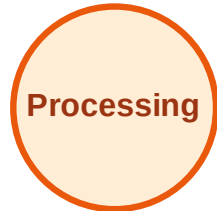
DFS Traversal

Step 17: No new neighbors from G

Legend



Complete



Processing



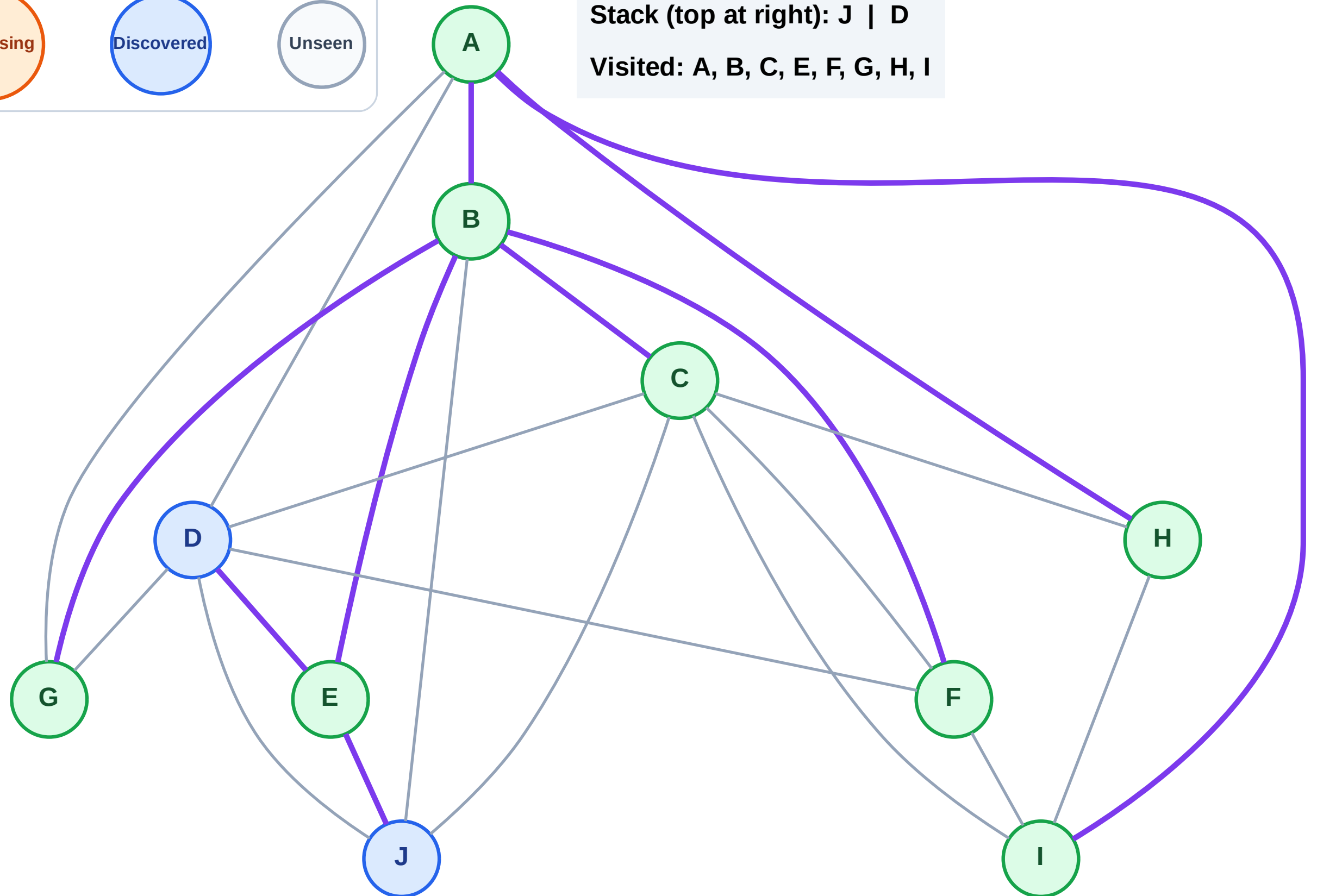
Discovered



Unseen

Stack (top at right): J | D

Visited: A, B, C, E, F, G, H, I



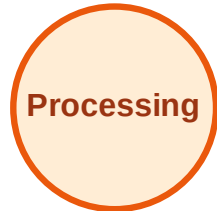
DFS Traversal

Step 18: Process node D

Legend



Complete



Processing



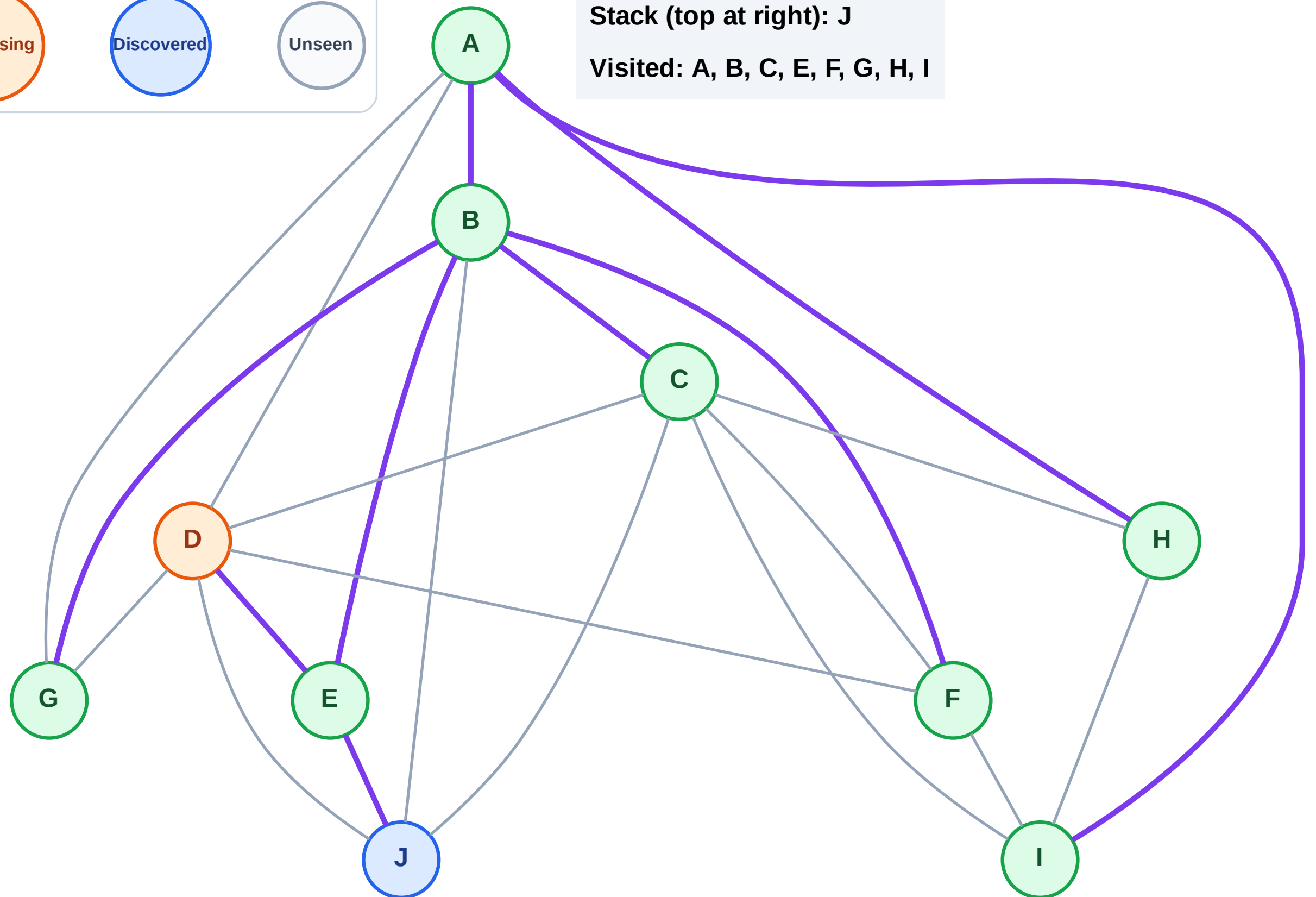
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, E, F, G, H, I



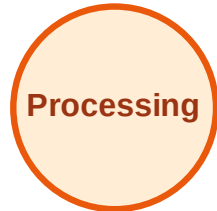
DFS Traversal

Step 19: No new neighbors from D

Legend



Complete



Processing



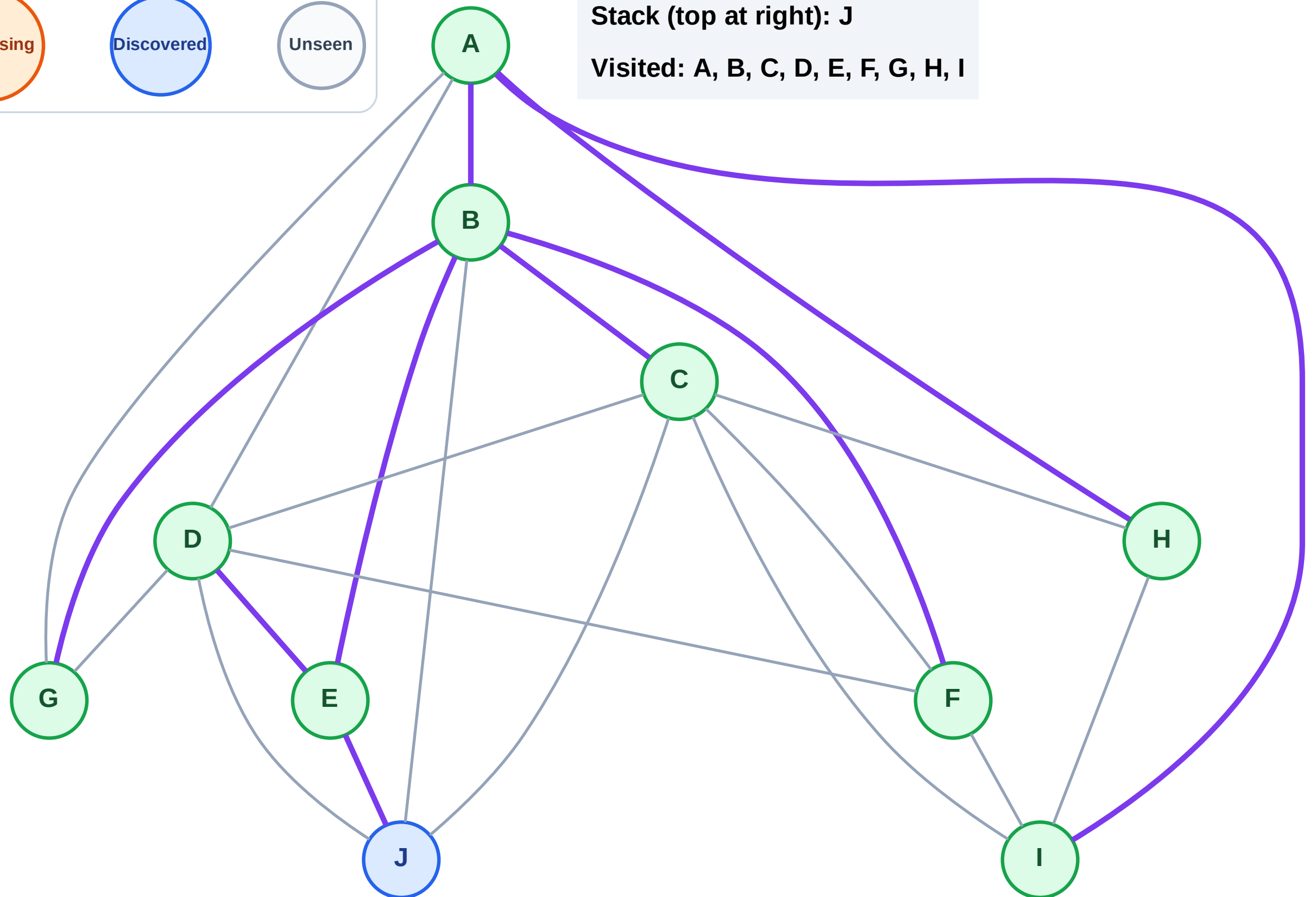
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

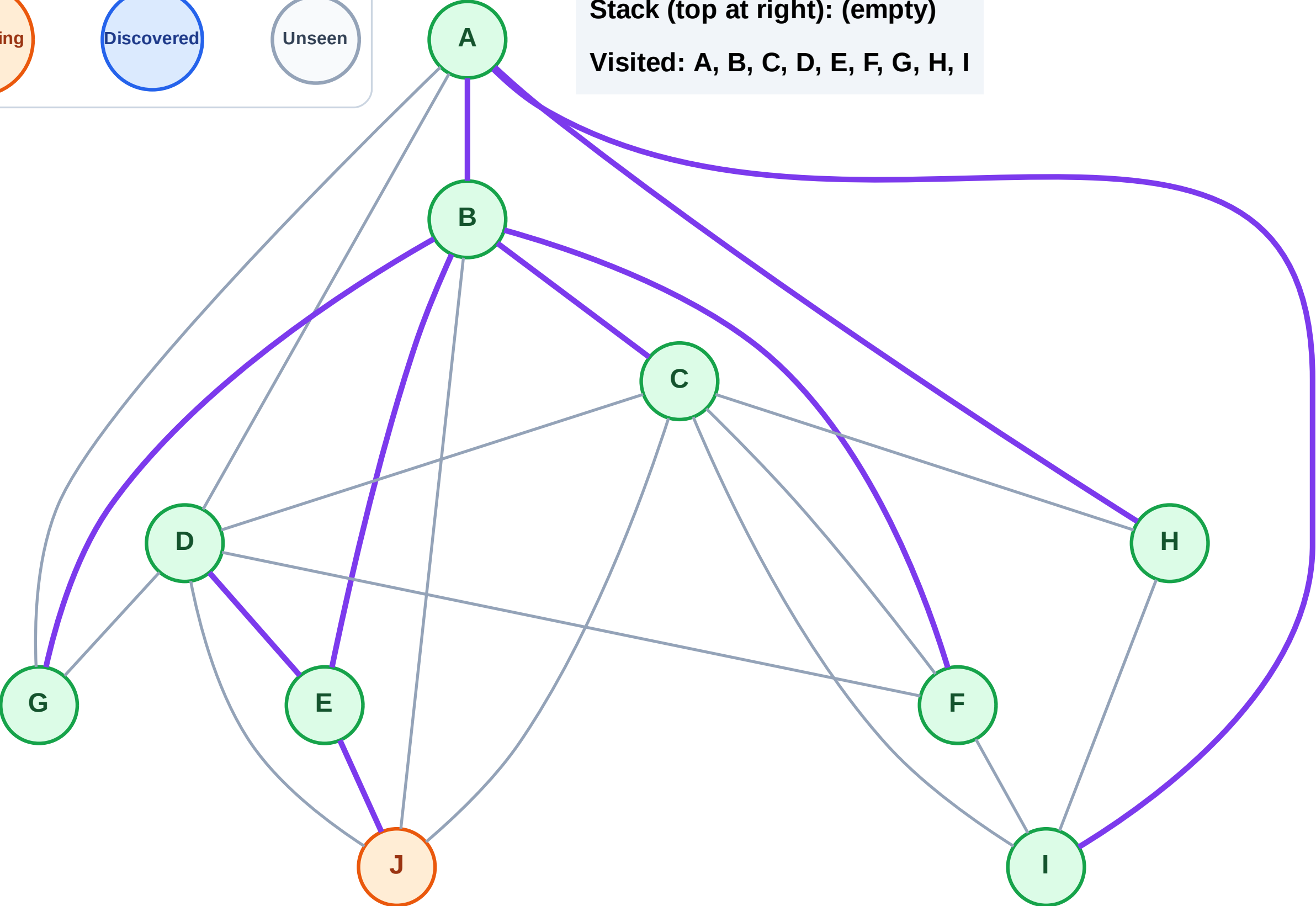
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



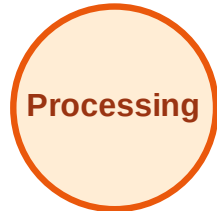
DFS Traversal

Step 21: No new neighbors from J

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

